

Quantity-Space Hyperprior Specification for Bayesian Hierarchical Models: Theory and Applications to Risk Preferences

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Abstract

Bayesian hierarchical models require hyperpriors that researchers rarely have direct intuition about. I develop a framework that recasts these choices in terms of interpretable predictive quantities. Rather than placing priors directly on structural parameters, the researcher specifies beliefs about observables such as certainty equivalents or choice probabilities, and a Jacobian correction maps those beliefs exactly into parameter space. A variance decomposition then clarifies how the prior distributes uncertainty between population-level variation and within-population heterogeneity. Because the prior is specified over observables rather than parameters, the same specification can be shared across structurally different models. The approach is illustrated with three rank-dependent utility models estimated on real lottery choice data, sharing one utility function but differing in their probability weighting functions and in the number of free parameters. All three models share a single quantity-space prior, making prior implications transparent and cross-model comparisons coherent. Because the shared prior fixes beliefs about observables across models, Bayes factors between the specifications measure evidence about functional form rather than differences in prior calibration. In the application the data decisively favor the two-parameter weighting functions over the one-parameter alternative, and favor the Prelec form within the two-parameter pair.

Keywords: Bayesian hierarchical models, hyperprior selection, prior elicitation, quantity-space priors, risk preferences, rank-dependent utility

JEL Classification: C11, C25, D81

1 Introduction

Bayesian hierarchical models (BHM) are becoming a standard tool for estimating heterogeneous preferences in experimental economics (Gao, Harrison and Tchernis 2022; Nilsson, Rieskamp and Wagenmakers 2011). These models jointly estimate individual-level parameters and the hyperparameters that govern their population distribution. By assuming exchangeability across subjects, BHM use the distribution of individual estimands to inform each subject’s prior, producing partially pooled estimates that balance individual data against population-level information. But this benefit creates an additional prior specification problem. Instead of specifying a prior for a single individual, the researcher must specify a *hyperprior* over the population distribution.

In many applications, each individual-level parameter is modeled with a two-parameter distribution, such as a Normal with a mean and variance. Each of those hyperparameters then receives its own prior, typically also parameterized by two values. A model with p individual-level parameters therefore often requires $4p$ hyperprior inputs. Common models of risk preferences under Expected Utility Theory (EUT) involve 2–3 parameters, models under RDU typically involve 3–5, and models under Cumulative Prospect Theory (CPT; Tversky and Kahneman 1992) can require up to 10 parameters.¹ As the dimensionality of the structural model grows, it becomes increasingly difficult to express domain knowledge

¹These ranges assume all models share a noise parameter and use utility and probability weighting functions. Two common utility functions are the CRRA and Expo-Power (Saha 1993), with 1 and 2 parameters respectively. Common probability weighting functions include the 2-parameter Prelec (1998) function, the 2-parameter Goldstein and Einhorn (1987) function, and the 1-parameter Power function (Quiggin 1982). CPT models utility and probability weighting separately over gains and losses, plus a loss aversion parameter, leading to 5–10 parameters depending on functional form choices.

directly in terms of these hyperprior parameters.

I develop a two-part framework for hyperprior specification in structural models. The first part concerns *quantity-space prior specification*. Researchers may have well-calibrated beliefs about predictive quantities, such as certainty equivalents (CEs) or choice probabilities, even when they have little intuition about structural parameters themselves. An injective mapping together with a Jacobian correction allows those beliefs to be translated into parameter-space priors without requiring the researcher to specify parameter-space beliefs directly.

The second part concerns the hierarchical problem itself. If individuals are exchangeable, then a prior for a randomly selected individual places restrictions on the hyperprior over the population. I show how these restrictions can be organized through a variance decomposition that separates uncertainty about the population mean from heterogeneity across individuals. In practice, this asks the researcher to specify beliefs about observables for a random individual and then to state how much of the resulting uncertainty should be attributed to heterogeneity in the population.

The contribution is therefore not to eliminate prior choices, but to move them into objects that are easier to interpret and defend. I extend the broader workflow of informed prior specification, in which priors are reasoned through their implications for predictive quantities, to hierarchical models of exchangeable populations. A further advantage is that the same quantity-space prior can be shared across structurally different models that make predictions about the same observables, enforcing consistency in beliefs about individual behavior independently of the latent model used to represent it.

I demonstrate the approach with three RDU models estimated on experimental

lottery-choice data from Harrison and Ng (2016), combining CRRA utility with Prelec, Goldstein-Einhorn, and Power probability weighting functions. The three models are chosen to exercise different aspects of the framework. All three share the same utility function but differ in their probability weighting, with the third using a simpler 1-parameter PWF that yields fewer free parameters and an overdetermined quantity-to-parameter mapping. By sharing a single quantity-space prior across all three, the application reveals where the models agree on predictive quantities (indicating that the structural specification is immaterial for those predictions) and where they diverge (indicating that the choice of functional form has substantive consequences). The shared prior also supports a formal comparison. Because all three models express identical prior beliefs about observables, Bayes factors between them measure evidence about functional form rather than differences in prior calibration.

The paper proceeds as follows. Section 2 reviews the literature on prior specification with emphasis on informed priors. Section 3 develops the general framework: first the quantity-space mapping and then the variance-decomposition approach to hyperprior construction. Section 4 applies the method to three RDU models estimated on experimental lottery-choice data. Section 5 presents the estimation results, together with the model comparison that the shared quantity-space prior makes possible. Section 6 discusses practical guidance, limitations, and extensions, and Section 7 concludes.

2 Literature: Prior Specification

Any proper prior yields a proper posterior under standard conditions, but that is a weak criterion for prior quality. A large literature therefore asks not

only whether a prior is admissible, but whether it is substantively and inferentially *good*.

For present purposes, that literature can be divided into two broad camps. The first concerns “default” priors for settings in which the researcher has little information about the model. The second concerns settings in which the researcher has at least some domain knowledge but may have difficulty expressing it in the model’s native parameterization. While the first camp has an extensive literature, largely focused on priors that do not distort inference in unintended ways,² the proposed method belongs squarely to the second camp.

The central idea in that second camp is to elicit beliefs in terms that are more natural than structural parameters. Kadane et al. (1980) established the paradigm of “predictive elicitation”, in which experts are asked about observable quantities and compatible parameter priors are derived from those predictive beliefs, an approach Kadane and Wolfson (1998) argue is cognitively more natural than direct elicitation of structural parameters. Schlag, Tremewan and van der Weele (2015), Garthwaite, Kadane and O’Hagan (2005), and Mikkola et al. (2024) review the practical elicitation methods developed since.

Once a researcher does have informed beliefs about relevant quantities, there remains the question of how to use them in Bayesian analysis. Gabry et al. (2019) propose a Bayesian workflow that iteratively searches for prior parameters producing reasonable prior predictive distributions of quantities about which the researcher has well-calibrated beliefs, and in the experimental economics literature Bland (2025) calibrates priors for structural models by examining their implications for certainty equivalents. Manderson and Goudie (2026) translate

²See Consonni, Fouskakis, Liseo and Ntzoufras (2018) for a comprehensive review.

predictive beliefs into prior distributions using Kullback-Leibler optimization, which can numerically match the prior predictive to a target distribution. Segert and Davis-Stober (2019) develop Jacobian-based transformations between parameter and observable spaces for individual-level models, with an emphasis on the conceptual interpretation of predictive distributions, and the same machinery is directly useful for informed priors. Relative to this literature, I contribute an exact Jacobian-based translation from quantity-space beliefs to parameter-space priors, together with a hierarchical extension that organizes the remaining hyperprior choices through a variance decomposition.

In the hierarchical setting specifically, Fuglstad et al. (2020) develop a framework for specifying joint priors on variance parameters by decomposing total variance along a tree structure, with Dirichlet or penalized complexity priors at each split. Their approach makes variance allocation interpretable for additive Gaussian models. I share that goal of making variance allocation an explicit researcher input and extend it in a different direction. Here the decomposition is expressed through two scalar parameters (α and κ) combined with closed-form moment-matching, which is convenient for hierarchies where the conditional variance may itself depend on the mean hyperparameter. In addition, the quantity-space mapping with Jacobian correction moves the elicitation problem from the model’s native parameters to predictive quantities, allowing the variance decomposition to be applied after beliefs about observables have been specified.

These contributions establish the foundation on which I build. What has been much less developed is how to extend predictive, quantity-based prior reasoning to hierarchical models, where the hyperprior specification problem is most acute. I provide that extension in analytic form.

3 General Hyperprior Specification from Predictive Quantities

While the paper’s main application is to structural models of risk preferences, the framework itself is more general. I therefore begin with a generic setup and then specialize it to the RDU application below.

3.1 Quantity-Space Prior Specification

Consider a model with p structural parameters $\boldsymbol{\theta} = (\theta^1, \dots, \theta^p) \in \Omega \subseteq \mathbb{R}^p$ and an injective mapping $\mathbf{g} : \Omega \rightarrow \mathbb{R}^k$ that takes parameters to $k \geq p$ predictive quantities $\boldsymbol{\vartheta} = \mathbf{g}(\boldsymbol{\theta})$.³ The researcher specifies a prior density $p_{\boldsymbol{\vartheta}}(\boldsymbol{\vartheta})$ directly in quantity space. The induced prior on the parameter space is

$$p_{\boldsymbol{\theta}}(\boldsymbol{\theta}) = p_{\boldsymbol{\vartheta}}(\boldsymbol{\vartheta}) \sqrt{\det(J_{\mathbf{g}}^{\top} J_{\mathbf{g}})} \quad (1)$$

where $J_{\mathbf{g}}$ is the $k \times p$ Jacobian matrix of $\mathbf{g}$, and $J_{\mathbf{g}}^{\top} J_{\mathbf{g}}$ is the $p \times p$ Gram matrix. The Gram determinant $\sqrt{\det(J_{\mathbf{g}}^{\top} J_{\mathbf{g}})}$ measures the p -dimensional volume of the parallelepiped spanned by the columns of J in $\mathbb{R}^k$.

This change-of-variables adjustment is well known, although it is usually presented in the special case $k = p$, where $\sqrt{\det(J_{\mathbf{g}}^{\top} J_{\mathbf{g}})} = |\det J|$. Segert and Davis-Stober (2019, pp. 108–110) discuss the detailed mathematical derivation of this formula and note the correction via the Coarea Formula needed when $k < p$. Given the focus of risk preference models, and the abundance of economically meaningful predictive quantities they generate, I focus on the case $k = p$ in the

³This is a weaker requirement than bijectivity, and permits a wider range of predictive quantities to be used in prior setting. The function $\mathbf{g}$ must be one-to-one on its domain Ω , but it does not need to map onto all of $\mathbb{R}^k$. The conditions needed for injectivity must be checked on a case by case basis.

main text, while noting that the same logic extends to $k > p$.⁴

For practical purposes, it is important that only the *forward* mapping $\mathbf{g}$ and its Jacobian are needed. The inverse $\mathbf{g}^{-1}$ never has to be solved explicitly. This matters because many economic quantities of interest do not have closed-form inverses, and numerical inversion can be costly even when $\mathbf{g}$ is injective.⁵ For example, the CE of a lottery under EUT with CRRA utility can be computed easily as a function of r , even though the inverse map from CE back to r has no closed-form solution.

Equation (1) is the starting point of the method. I assume that the researcher can specify a proper prior over $k \geq p$ predictive quantities for a randomly selected individual from a population of exchangeable individuals, and that these quantities are related to the p structural parameters through a provably injective map $\mathbf{g}$. I refer to this individual-level quantity prior as the “direct prior”. For a model of a single random individual, the direct prior completes the prior specification exercise. For a BHM, however, it remains to construct a hyperprior that is compatible with it.

3.2 *Hyperprior Construction via Variance Decomposition*

I begin with the scalar case $k = p = 1$, where the logic is easiest to see. Let ϑ_i be a scalar predictive quantity for individual i , and suppose the researcher

⁴When $k > p$, the image of $\mathbf{g}$ is a p -dimensional submanifold of $\mathbb{R}^k$, and the researcher’s quantity-space density should be understood as evaluated on this image rather than as a free-standing density on all of $\mathbb{R}^k$. In this case, the induced prior (1) is not automatically normalized, but Bayesian inference requires only $p_\theta \propto p_\vartheta(\mathbf{g}(\boldsymbol{\theta}))\sqrt{\det(J^\top J)}$. See Segert and Davis-Stober (2019, pp. 108–110) for the measure-theoretic details.

⁵In practice, tools like the SymPy python library or Mathematica can be used to derive an analytical form for the Jacobian, if one exists. Additionally, automatic differentiation, as is used in the popular JAX python library, efficiently computes both $\mathbf{g}(\boldsymbol{\theta})$ and $J_{\mathbf{g}}$ directly without first having to solve their analytical form, further reducing the barriers for use of this method. Proving injectivity, however, can be challenging, as we address later.

has specified a direct prior $p_{\vartheta}^*(\vartheta)$ with mean m^* and variance v^* . This prior represents beliefs about the quantity for a single randomly selected individual from the population. If individuals are exchangeable, then that direct prior must integrate over all possible heterogeneity in the population:

$$p_{\vartheta}^*(\vartheta \mid m^*, v^*) = \int p_{\vartheta}(\vartheta \mid \gamma) p(\gamma \mid \xi) d\gamma \quad (2)$$

where m^* and v^* are the mean and variance of the direct prior, γ is a vector of hyperparameters governing the population distribution of ϑ , and $p(\gamma \mid \xi)$ is the hyperprior over those hyperparameters. Equation (2) is the ideal restriction: it says that the direct prior for a random individual should be recoverable by integrating over the population hierarchy.

When p_{ϑ}^* is itself parametric, this equality is generally not achievable exactly. The right-hand side is a *compound distribution*, formed by integrating the individual-level density over the hyperprior. For most families, that compound distribution is not in the same family as the direct prior. A compound Normal is not Normal, and a compound Beta is not Beta. So in general no choice of ξ makes the right-hand side of (2) exactly equal to the parametric direct prior on the left.

A practical response is to approximate (2) by matching the first two moments of the hierarchical predictive distribution to m^* and v^* .⁶ Any parameterization of the individual-level distribution implies a conditional mean and variance that are functions of the hyperparameters. I denote these by $\gamma_m := \mathbb{E}[\vartheta_i \mid \gamma]$ and $\gamma_v := \text{Var}[\vartheta_i \mid \gamma]$, and formulate the decomposition in terms of these moment

⁶The method requires moments to *exist*, not to be available in closed form. Distributions whose moments must be computed numerically, such as the Logit-Normal, are perfectly feasible. Unfortunately, this does rule out the popular Cauchy distribution, which has *no* moments. Instead of matching moments, this exercise could be rearranged to instead match quantiles, which would allow for the Cauchy to be used.

hyperparameters.⁷

Each moment hyperparameter has a two-parameter hyperprior whose parameters biject to its first two moments, so ξ can be represented by four unknown moments. I write ξ_{mm} and ξ_{mv} for the mean and variance of the hyperprior over γ_m , and ξ_{vm} and ξ_{vv} for the mean and variance of the hyperprior over γ_v . The laws of iterated expectations and total variance then imply

$$m^* = \mathbb{E}[\vartheta_i] = \mathbb{E}[\gamma_m \mid \xi_{mm}, \xi_{mv}] \quad (3)$$

$$v^* = \text{Var}[\vartheta_i] = \mathbb{E}[\gamma_v \mid \xi_{vm}, \xi_{vv}] + \text{Var}[\gamma_m \mid \xi_{mm}, \xi_{mv}] \quad (4)$$

Equations (3)–(4) hold for any hyperprior family with defined first and second moments. They are not tied to a particular distributional choice. At this stage they provide two of the four constraints needed to determine ξ .

The remaining two constraints come from decomposing v^* into components that have a direct interpretation. Equation (4) separates total uncertainty into expected within-population heterogeneity ($\mathbb{E}[\gamma_v \mid \xi_{vm}, \xi_{vv}]$) and uncertainty about the population mean ($\text{Var}[\gamma_m \mid \xi_{mm}, \xi_{mv}]$). I parameterize this partition with the ratio

$$\alpha v^* = \text{Var}[\gamma_m \mid \xi_{mm}, \xi_{mv}] \quad (5)$$

$$(1 - \alpha) v^* = \mathbb{E}[\gamma_v \mid \xi_{vm}, \xi_{vv}] \quad (6)$$

where $\alpha \in (0, 1)$ is the variance partition ratio, chosen by the researcher.⁸

⁷For some families, such as the Normal, the mean and variance are the hyperparameters themselves. For others, such as the Beta, they are functions of multiple hyperparameters, and the translation from moments to the actual hyperprior parameters requires additional algebra. Appendix A works out this translation for three common families.

⁸The ratio α is numerically equivalent to the variance partition coefficient (Goldstein, Browne and Rasbash 2002) and the intraclass correlation coefficient, which are widely used

Because α is a ratio of variances, it is unit-free. The same value of α therefore expresses the same structural belief about the balance between population-mean uncertainty and individual heterogeneity regardless of the units of the quantity being modeled. Low α reflects a belief in substantial heterogeneity; high α reflects a belief that most uncertainty comes from the location of the population mean.

Equations (3) and (5) together provide two moment conditions that uniquely determine ξ_{mm} and ξ_{mv} :

$$\mathbb{E}[\gamma_m \mid \xi_{mm}, \xi_{mv}] = m^* \quad (7)$$

$$\text{Var}[\gamma_m \mid \xi_{mm}, \xi_{mv}] = \alpha v^* \quad (8)$$

These are known quantities, and the hyperprior family is assumed bijective in its first two moments, so this system has a unique solution ξ_{mm} and ξ_{mv} .

With α , m^* , and v^* specified, the remaining hierarchical choice concerns how uncertain the researcher is about the heterogeneity level. To parameterize this in a unit-free way, I introduce the coefficient of variation of heterogeneity:

$$\begin{aligned} \kappa &:= \frac{\text{SD}[\gamma_v \mid \xi_{vm}, \xi_{vv}]}{\mathbb{E}[\gamma_v \mid \xi_{vm}, \xi_{vv}]} \\ &= \frac{\text{SD}[\gamma_v \mid \xi_{vm}, \xi_{vv}]}{(1 - \alpha) v^*} \end{aligned} \quad (9)$$

which measures relative uncertainty about the variance hyperparameter. As $\kappa \rightarrow 0$, the researcher is expressing near-certainty that population heterogeneity is close to its target level $(1 - \alpha)v^*$. Larger values of κ represent greater uncertainty.

as posterior diagnostics in multilevel modelling. Fuglstad et al. (2020) also use variance proportions as prior inputs, but they partition variance across model components, with the proportions treated as random variables. In the proposed framework, this ratio is an input specified by the researcher, and is not estimated.

At $\kappa = 1$ the standard deviation of the variance equals its mean, and $\kappa > 1$ indicates very diffuse beliefs about heterogeneity. In other words, α sets the heterogeneity target and κ states how strongly that target is held. Equations (6) and (9) then provide two moment conditions that uniquely determine ξ_{vm} and ξ_{vv} :

$$\mathbb{E}[\gamma_v \mid \xi_{vm}, \xi_{vv}] = (1 - \alpha) v^* \quad (10)$$

$$\text{Var}[\gamma_v \mid \xi_{vm}, \xi_{vv}] = \kappa^2 (1 - \alpha)^2 (v^*)^2 \quad (11)$$

As in (7), the right-hand side of (10) is composed of researcher-specified constants, and the hyperprior over γ_v is presumed to have a bijective mapping from its parameters to its first two moments, so (10) has a unique solution ξ_{vm} and ξ_{vv} . Thus, all four hyperprior parameters $(\xi_{mm}, \xi_{mv}, \xi_{vm}, \xi_{vv})$ are now determined. The researcher's beliefs about heterogeneity are fully characterized by α , which controls the level of heterogeneity relative to population-mean uncertainty, and κ , which controls confidence in that level.

The resulting hierarchical predictive matches the direct prior exactly in its first two moments by construction. Higher moments, such as skewness, kurtosis, and tail probabilities, will generally differ because the compound distribution produced by the hierarchy is not in the same family as the direct prior. The size of this discrepancy depends in part on α and κ . As both approach zero, the hyperprior concentrates and the compound distribution collapses toward the individual-level distribution with mean m^* and variance v^* . If the direct prior uses the same family as the individual-level distribution, it is recovered exactly in this limit. That is not required by the construction, but it is convenient and is the approach used in the application below. For nonzero α and κ , the discrepancy

is typically small, but it should be checked by comparing the hierarchical predictive to the direct prior. Appendix A provides this comparison for three standard families.

3.3 *Family-Specific Translation and Feasibility*

The moment conditions (7)–(10) can be solved in closed form for standard families, but the translation from (α, κ) to actual hyperprior parameters is family-specific. In the Normal-Inverse-Gamma hierarchy, the conditional variance is a single parameter σ^2 with its own independent hyperprior, so κ maps directly to the Inverse-Gamma shape: $\alpha_0 = 2 + 1/\kappa^2$. In the Beta-Beta hierarchy used in the application below, the conditional variance $\mu(1 - \mu)\psi$ depends on both hyperparameters, so the translation requires cross-moment calculations. Appendix A provides the full derivations, graphical calibrations, and approximation analysis for three families: Normal-Inverse-Gamma (support on $\mathbb{R}$), Beta-Beta (support on $[0, 1]$), and Gamma-Inverse-Gamma (support on $(0, \infty)$).

The feasibility region in (α, κ) space is also family-dependent. Families where the conditional variance is a single parameter (NIG, Gamma-Inverse-Gamma) admit all $\kappa > 0$ with no joint constraint on α . Families where the conditional variance depends on the mean hyperparameter (Beta-Beta) impose both a lower bound on κ (from the irreducible conditional variance variation contributed by the mean hyperparameter) and an upper bound (from the requirement that the dispersion hyperprior remain proper). In all cases, the feasibility constraints are computable in closed form from the researcher’s direct prior and can be checked before estimation.

3.4 *Practical Interpretation*

With k predictive quantities, the decomposition applies separately to each quantity. Each quantity requires a direct prior mean m_k^* and variance v_k^* , a variance partition ratio α_k , and a heterogeneity confidence κ_k , yielding $4k$ researcher inputs in total. The value of the framework is therefore not parsimony in the number of choices, but a change in what those choices mean. The hyperprior predictive matches the direct prior in its first two moments by construction, making the calibration more direct (Bland 2025; Gabry et al. 2019), and the remaining hierarchical choices are interpretable, with α governing how much uncertainty is attributed to heterogeneity across individuals rather than to uncertainty about the population mean and κ governing how tightly that heterogeneity level is held.

The practical workflow is therefore two-step. First, specify a quantity-space prior for a randomly selected individual using predictive quantities that admit an injective map from parameter space. Second, specify beliefs about how the resulting uncertainty should be allocated across the hierarchy. I now demonstrate this workflow for three RDU models estimated on experimental lottery-choice data, where the predictive quantities are certainty equivalents and a choice probability.

4 **Application: RDU with Experimental Lottery-Choice Data**

Researchers studying risk preferences face a practical problem: the choice between alternative utility functions and probability weighting functions is often

made before estimation, and each candidate model requires its own hyperprior specification. When those specifications are made in parameter space, switching from one model to another means starting the hyperprior exercise from scratch, because the parameters have no common meaning across models. The quantity-space approach avoids this problem. The researcher specifies beliefs about observables once, and those beliefs translate into parameter-space priors for any structural model that maps to the same quantities. This section demonstrates that property by applying the framework to three RDU models that differ only in the functional forms entering the Jacobian of the forward map.

4.1 *RDU Framework*

Under RDU, the value of a lottery with n outcomes $x_1 \leq \dots \leq x_n$ and associated probabilities $p_1, \dots, p_n$ is

$$U = \sum_{i=1}^n d_i u(x_i) \quad (12)$$

where $u(\cdot)$ is a utility function and d_i are decision weights derived from a probability weighting function (PWF) w . The decision weights are constructed from the decumulative probabilities $S_i = \sum_{j=i}^n p_j$ as

$$d_i = w(S_i) - w(S_{i+1}) \quad (13)$$

where $w(S_{n+1}) = 0$. The probability that lottery A is chosen over B follows the logistic choice function with Contextual Utility normalization (Wilcox 2011):

$$P(A) = \frac{1}{1 + \exp(-\Delta)}, \quad (14)$$

$$\Delta = \frac{U(A) - U(B)}{\lambda(u(\bar{x}) - u(\underline{x}))}$$

where $\lambda > 0$ is a Fechnerian noise scaling parameter and $u(\bar{x}) - u(\underline{x})$ is the utility range across all outcomes with positive probability in both lotteries. This choice model is common to all three specifications below.

4.2 Three RDU Specifications

I consider three combinations of utility function and PWF that vary in flexibility and functional form. The first follows Gao, Harrison and Tchernis (2022) and is the workhorse specification in the experimental economics literature. The second retains the same CRRA utility but replaces the PWF with a different 2-parameter form, isolating the effect of the probability weighting specification on posterior predictions. The third uses a simpler 1-parameter PWF with fewer free parameters, creating an over-determined ($k > p$) case that exercises the Gram determinant machinery from Section 3.

Model 1: CRRA + Prelec

The first specification uses the CRRA utility function and the 2-parameter Prelec (1998) PWF:

$$u(x) = \frac{x^{1-r}}{1-r} \tag{15}$$

$$w(p) = \exp\left(-\eta(-\ln p)^\phi\right) \tag{16}$$

where r governs risk aversion (with $r = 0$ corresponding to risk neutrality), $\phi > 0$ controls the curvature of the PWF, and $\eta > 0$ controls its elevation. When $\phi = \eta = 1$ the PWF is linear, reducing to EUT. The model has 4 parameters: (r, ϕ, η, λ) .

This specification is a workhorse of structural estimation of risk preferences.

Andersen, Harrison, Lau and Rutström (2008) use it for joint estimation of risk and time preferences, Bruhin, Fehr-Duda and Epper (2010) to uncover heterogeneity in probability distortion, and Tanaka, Camerer and Nguyen (2010) in field experiments in developing countries. von Gaudecker, van Soest and Wengström (2011) characterize risk attitudes in a broad population with it, and Gao, Harrison and Tchernis (2022) build it into a hierarchical Bayesian welfare analysis.

Model 2: CRRA + Goldstein-Einhorn

The second specification retains the CRRA utility function and pairs it with the 2-parameter Goldstein and Einhorn (1987) PWF:

$$u(x) = \frac{x^{1-r}}{1-r} \tag{17}$$

$$w(p) = \frac{\delta p^\gamma}{\delta p^\gamma + (1-p)^\gamma} \tag{18}$$

where r governs risk aversion as in Model 1, $\gamma > 0$ controls the curvature (discriminability) of the PWF, and $\delta > 0$ controls its elevation (attractiveness). When $\gamma < 1$ and $\delta = 1$ the function produces the empirically well-documented inverse-S shape, overweighting small probabilities and underweighting large ones. When $\gamma = 1$ and $\delta = 1$ the PWF reduces to the identity, recovering EUT. Gonzalez and Wu (1999) give the two parameters a psychological interpretation, with γ measuring how sharply an individual discriminates among probabilities and δ how attractive the gamble is overall, and the function has a long record in experimental work (Lattimore, Baker and Witte 1992; Stott 2006). Pairing it with the same CRRA utility as Model 1 isolates the weighting specification,

since any difference between the two models' predictions comes from the functional form of w rather than from utility. The model has 4 parameters: $(r, \gamma, \delta, \lambda)$.

Model 3: CRRA + Power

The third specification pairs the CRRA utility function with the 1-parameter Power PWF (Quiggin 1982):

$$u(x) = \frac{x^{1-r}}{1-r} \quad (19)$$

$$w(p) = p^\delta \quad (20)$$

where $\delta > 0$ controls the curvature of the PWF. When $\delta = 1$ the PWF is linear, reducing to EUT. Unlike the Prelec and Goldstein-Einhorn functions, the Power PWF does not produce an inverse-S shape: for $\delta < 1$ it uniformly overweights all probabilities, and for $\delta > 1$ it uniformly underweights them. The model has 3 structural parameters: (r, δ, λ) .

This specification serves two purposes. First, it shares the CRRA utility with Models 1 and 2, so any CE difference between Models 1, 2, and 3 is attributable entirely to the PWF. The Power PWF is nested within the Prelec family (setting $\phi = 1$ in (16) gives $w(p) = p^\eta$, so the Power exponent δ plays the role of η), making Models 1 and 3 nested, which supports the formal nested comparison summarized in Section 5.4. Second, with only 3 structural parameters mapped to 4 predictive quantities, it is an over-determined case ($k = 4 > p = 3$). The change-of-variables formula uses the Gram determinant $\sqrt{\det(J_{\mathbf{g}}^\top J_{\mathbf{g}})}$ in place of $|\det J_{\mathbf{g}}|$, as described in Section 3. This demonstrates that the framework handles the $k > p$ case with no additional machinery beyond the Jacobian

computation.

Together the three specifications cover both the square case, with Models 1 and 2 at $k = p = 4$, and the over-determined case, with Model 3 at $k = 4 > p = 3$.

4.3 *Quantity-Space Mapping for RDU*

Under each specification, each individual has 3 or 4 structural parameters θ , and a traditional parameter-space approach would require setting 12–16 hyperprior parameters. Following the method in Section 3, I instead specify 4 direct priors on predictive quantities about which a researcher has well-calibrated beliefs: 3 certainty equivalents (CEs) and one choice probability. The key observation is that these quantities are the same regardless of which utility function or PWF generates them.

The CE is the certain amount of money that provides the same utility as a lottery, translating structural parameters into dollar-denominated quantities about which researchers have direct intuition (Abdellaoui 2000). A researcher who has no feel for plausible values of Prelec’s elevation parameter η can still say whether a typical experimental subject would trade a lottery paying \$50 with probability 0.25 and \$10 otherwise for its expected value of \$20 with certainty. CEs have been widely used for post-estimation welfare analysis (Gao, Harrison and Tchernis 2022; Harrison and Ng 2016; Monroe 2023) and prior calibration (Bland 2025).

For any RDU specification, the preference parameters θ determine the CE of a lottery through

$$u(\text{CE}; \theta) = \sum_j d_j(\theta) u(x_j; \theta) \quad (21)$$

where $u(\cdot; \boldsymbol{\theta})$ is the utility function and $d_j(\boldsymbol{\theta})$ are decision weights. Choosing three reference lotteries L_1, L_2, L_3 with suitably diverse characteristics defines a mapping from preference parameters to CEs. For Model 1 this is $\mathbf{g}_1(r, \phi, \eta) = (\text{CE}_1, \text{CE}_2, \text{CE}_3)$; for Model 2 it is $\mathbf{g}_2(r, \gamma, \delta) = (\text{CE}_1, \text{CE}_2, \text{CE}_3)$; for Model 3 it is $\mathbf{g}_3(r, \delta) = (\text{CE}_1, \text{CE}_2, \text{CE}_3)$, where the mapping has 3 outputs from only 2 preference parameters. In each case the noise parameter λ does not affect CEs but governs choice precision. Conditioning on the preference parameters and evaluating $P(A)$ from (14) for a reference lottery pair (A, B) pins down λ , extending the mapping to 4 quantities. For Models 1 and 2 ($k = p = 4$), the Jacobian $J_{\mathbf{g}}$ is square and the induced prior uses $|\det J_{\mathbf{g}}|$. For Model 3 ($k = 4 > p = 3$), $J_{\mathbf{g}}$ is 4×3 and the induced prior uses the Gram determinant $\sqrt{\det(J_{\mathbf{g}}^T J_{\mathbf{g}})}$, as derived in Section 3.⁹

The crucial point is that the CEs and choice probability are *model-independent observables*: they describe behavior rather than parameters. The direct prior over $(\text{CE}_1, \text{CE}_2, \text{CE}_3, P)$ is therefore the same for all three models. What differs is only the Jacobian $J_{\mathbf{g}}$, which depends on the functional forms of u and w . The quantity-space prior specification is done once, and the change-of-variables formula translates it into a different parameter-space prior for each model.

All four predictive quantities are bounded: CEs by lottery outcomes and

⁹The change-of-variables formula requires that the mapping $\mathbf{g}$ be injective on the support of the prior. A nonvanishing Jacobian (or Gram) determinant establishes only *local* invertibility and is necessary but not sufficient. Proving injectivity for the mapping is nontrivial and must be assessed on a case-by-case basis. Appendix B provides an analytic proof for the CRRA + Prelec model. The same proof structure extends to the CRRA + Goldstein-Einhorn model, since both use CRRA utility and a 2-parameter PWF satisfying the same two-point injectivity conditions (see Section B.2.1 in Appendix B). For the CRRA + Power model, the proof extends directly: the Power PWF is strictly monotone in δ , so a single decision weight identifies δ , and the remaining CE identifies r through the same CRRA monotonicity argument. Numerical verification via the Gram determinant on dense grids confirms full column rank of the Jacobian.

the choice probability by $[0, 1]$. Each CE can be rescaled to $(0, 1)$ by the linear transformation $z = (Y - \underline{x})/(\bar{x} - \underline{x})$, so the natural population hierarchy uses Beta distributions throughout. Let $z_i^k \in (0, 1)$ denote the normalized value of quantity k for individual i . The hierarchical model is:

$$z_i^k \mid \mu_k, \psi_k \sim \text{Beta}\left(\mu_k \frac{1-\psi_k}{\psi_k}, (1 - \mu_k) \frac{1-\psi_k}{\psi_k}\right) \quad (22)$$

$$\mu_k \sim \text{Beta}(a_\mu^k, b_\mu^k) \quad (23)$$

$$\psi_k \sim \text{Beta}(a_\psi^k, b_\psi^k) \quad (24)$$

where $\mu_k \in (0, 1)$ is a hyperparameter governing the population mean of quantity k and $\psi_k \in (0, 1)$ is a hyperparameter governing dispersion, which together determine the conditional variance:

$$\text{Var}[z_i^k \mid \mu_k, \psi_k] = \mu_k(1 - \mu_k)\psi_k \quad (25)$$

The mean-dispersion parameterization allows independent hyperpriors on μ_k and ψ_k , avoiding the conditional domain restriction that would arise from parameterizing in terms of mean and variance directly.

The hierarchical model has 4 hyperprior parameters per quantity. The variance decomposition of Section 3.2 replaces these with 4 interpretable inputs per quantity: a direct prior mean m_k^* and variance v_k^* , a variance partition ratio α_k , and a heterogeneity confidence κ_k . For the Beta-Beta hierarchy, this translation is not as simple as in the NIG case because the conditional variance depends on both μ_k and ψ_k , so the mapping requires cross-moment calculations. The complete derivation, including the feasibility region and κ bounds, is given in Appendix A.3. Because α_k and κ_k are unit-free ratios, it is reasonable in this

application to share them across quantities. Appendix D shows that this CE-space specification induces a sensible prior in parameter space, centered near mild risk aversion with approximately linear probability weighting, without requiring the researcher to specify beliefs about structural parameters directly.

5 Estimation and Results

I estimate the model using lottery choice data from Harrison and Ng (2016), in which 111 subjects each made 80 binary choices between three-outcome lotteries with prize amounts of \$10, \$30, and \$50. All three RDU models are estimated on the same data, each from 80 choices per subject, a setting where the hyperprior has real influence, making it a useful testbed for the proposed framework.¹⁰ I now describe the specific choices made, including the reference lottery design, direct prior beliefs, and variance decomposition parameters.

The quantity-space framework of Section 4.3 requires reference lotteries for the CE mapping and a reference pair for the choice probability, chosen so that the map from parameters to predictive quantities is injective. Injectivity must be verified for each specification separately, and Appendix B does so for all three. The analytic proof rests on a two-point argument that covers both the Prelec and Goldstein-Einhorn forms, the Power form follows from monotonicity in its single parameter, and a numerical rank check of the Gram determinant supplements the argument for Model 3. The design uses two binary lotteries with distinct probabilities on the higher outcome and a third binary lottery with shared probability $p_3 = p_1$ whose outcome ratio differs from that of L_1 . No pair

¹⁰MCMC estimation details, including chain counts, convergence diagnostics, and acceptance rates for each model, are reported in Appendix E.

of lotteries with overlapping outcomes exhibits first-order stochastic dominance, which avoids obvious ordering contradictions when specifying independent marginal distributions over the CE quantities. Table 1 presents a set of lotteries satisfying these conditions, with the pair (L_1, L_3) chosen for the choice probability mapping.

Lottery	Outcomes	Probability on better outcome	Expected Value
L_1	(\$10, \$50)	$p_1 = 0.25$	\$20
L_2	(\$20, \$40)	$p_2 = 0.5$	\$30
L_3	(\$12, \$44)	$p_3 = 0.25$	\$20

Table 1: Reference lottery design for prior specification.

For each quantity, the researcher must specify beliefs about what that quantity might be for a randomly selected individual. The variance decomposition of Section 3.2 requires a direct prior mean and variance, but there are many ways to elicit beliefs about predictive quantities that yield a proper direct prior. One natural approach for this application is to specify a mean and the probability that a randomly selected individual is risk-seeking. For CEs, risk-seeking means having a CE above the expected value. For the choice probability, it means preferring the riskier of two equal-EV lotteries. For a Beta distribution with fixed mean, this tail probability uniquely determines the concentration and therefore the variance.

The reference pair for the choice probability is L_1 and L_3 , which share the same expected value but differ in spread: L_1 has outcomes \$10 and \$50 while L_3 has \$12 and \$44. Since L_3 is a mean-preserving contraction of L_1 , preferring L_1 is risk-seeking in the Rothschild and Stiglitz (1970) sense, regardless of the underlying preference model. The choice probability P represents the probability that L_1 is chosen over L_3 , so $P > 0.5$ indicates risk-seeking.

The following calibration is illustrative, reflecting broad empirical regularities from the experimental risk preference literature rather than a specific pilot study. I specify that the mean CE for each lottery is 85% of the distance from the lower bound to the expected value, reflecting a belief that the typical individual is mildly risk-averse, with a 15% probability of being risk-seeking. The same 15% tail probability applies to the choice probability: $P(P > 0.5) = 0.15$, with a mean of 0.40, reflecting the belief that most individuals are risk-averse.¹¹ The mean and tail probability are sufficient to uniquely determine the direct prior variance for each quantity, as shown in Table 2.

Quantity	Mean	Reference	$P(\text{risk-seeking})$	Implied Var.	Implied SD
CE ₁	\$18.5	EV ₁ = \$20	0.15	2.08	\$1.44
CE ₂	\$28.5	EV ₂ = \$30	0.15	2.05	\$1.43
CE ₃	\$18.8	EV ₃ = \$20	0.15	1.33	\$1.15
P	0.40	0.50	0.15	0.0090	0.0946

Table 2: Direct prior beliefs for the four predictive quantities. CE rows are in dollars (variances in squared dollars); the P row is on the probability scale.

Table 2 specifies the resulting direct prior for each quantity. The variance decomposition of Section 3.2 splits each direct prior variance into population-mean uncertainty and within-population heterogeneity, governed by α_k and κ_k . Because α_k is a variance ratio and κ_k is a coefficient of variation, both are unit-free and can reasonably be shared across quantities with different natural units. I set $\alpha_k = 0.2$ for all quantities, meaning that 20% of the direct prior variance is attributed to uncertainty about the population mean and the remaining 80% to heterogeneity across individuals. I set $\kappa_k = 0.25$, so the coefficient of variation of

¹¹The common tail probability across all four quantities is a pragmatic simplification. In applications where lottery-specific beliefs differ, quantity-specific values can be used without modification to the framework.

the within-population variance is 25%. This reflects moderately high confidence in the heterogeneity specification, meaning the prior places most of its mass near the target level of heterogeneity while still permitting the data to move it.¹²

The quantity-space hierarchy defines a prior on $(\text{CE}_1, \text{CE}_2, \text{CE}_3, P)$. The change-of-variables formula transforms this to the parameter-space prior used in sampling. For Models 1 and 2 ($k = p = 4$):

$$p(r, \phi, \eta, \lambda) = p_Y(\mathbf{g}_1(r, \phi, \eta, \lambda)) \left| \det J_{\mathbf{g}_1}(r, \phi, \eta, \lambda) \right| \quad (26)$$

$$p(r, \gamma, \delta, \lambda) = p_Y(\mathbf{g}_2(r, \gamma, \delta, \lambda)) \left| \det J_{\mathbf{g}_2}(r, \gamma, \delta, \lambda) \right| \quad (27)$$

For Model 3 ($k = 4 > p = 3$), the Jacobian is non-square and the induced prior uses the Gram determinant:

$$p(r, \delta, \lambda) = p_Y(\mathbf{g}_3(r, \delta, \lambda)) \sqrt{\det(J_{\mathbf{g}_3}^\top J_{\mathbf{g}_3})} \quad (28)$$

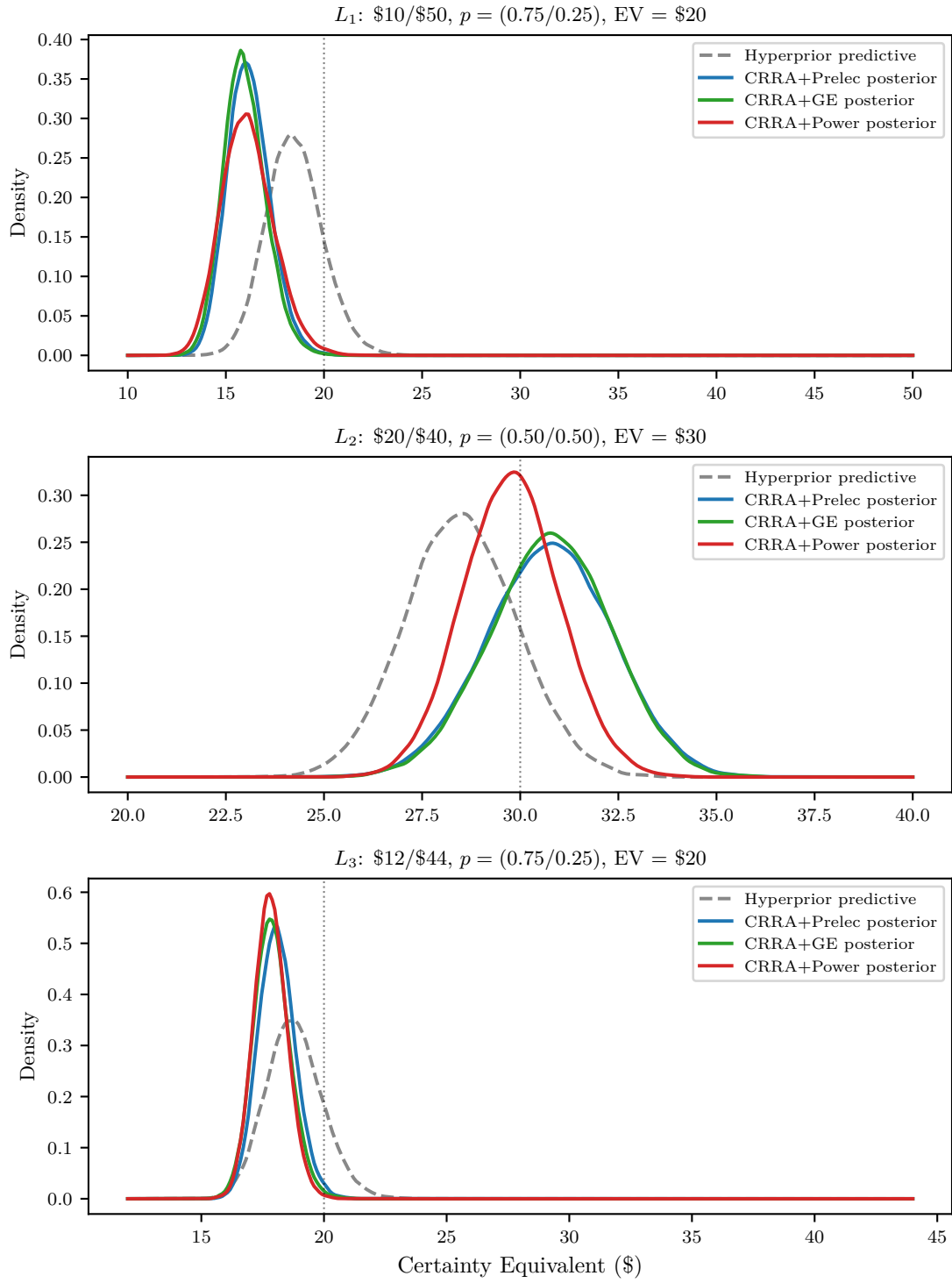
The quantity-space density p_Y is identical in all three equations. Only the forward map $\mathbf{g}$ and its Jacobian differ. All three models share the same prior beliefs about certainty equivalents and choice probabilities, while inducing different priors over their respective structural parameters.

5.1 Cross-Model Posterior Predictive Certainty Equivalents

I begin the discussion of results with the population-level predictive distributions, which show what each estimated model implies for a new, unobserved individual. Figure 1 shows the hyper-posterior predictive distribution of certainty equivalents for each of the three reference lotteries under all three models, together with the

¹²Appendix A.3 illustrates how these choices control the hierarchical structure and confirms that the moment-matched hierarchical predictive closely tracks the direct prior.

Figure 1: Hyper-posterior predictive CE distributions across three models



shared hyperprior predictive.¹³

The estimation for each model starts from identical beliefs about these quantities, so differences in the posterior predictive CEs reflect the interaction between the structural model’s Jacobian and the data likelihood. For lotteries L_1 and L_3 , which place probability 0.75 on the worse outcome, I find closely aligned CE posteriors across the three models, with median CEs of \$16.1, \$15.9, and \$16.1 for L_1 under Models 1, 2, and 3 respectively. All three are concentrated below the expected value, indicating that the estimated population is risk-averse for these asymmetric lotteries. Model 3, which has only 3 free parameters mapped to 4 predictive quantities through the Gram determinant, produces posterior CEs within \$0.25 of the 4-parameter models on these lotteries.

For lottery L_2 , which splits probability equally between outcomes, a more nuanced pattern emerges. The Prelec and Goldstein-Einhorn posteriors (Models 1 and 2) are nearly identical, with median CEs of \$30.8 and \$30.8 respectively. The Power posterior (Model 3) is lower at \$29.8. Since all three models share the CRRA utility function, these differences are attributable entirely to the probability weighting specification. The agreement between Prelec and Goldstein-Einhorn is notable because these are non-nested functional forms. Both can produce inverse-S shaped distortions, and the reference lotteries use moderate probabilities ($p = 0.25$ and $p = 0.50$) where the two functions can closely approximate each other by adjusting their curvature and elevation parameters. The two forms differ most in how they weight extreme probabilities near 0 or 1, so experimental designs with highly skewed lotteries would be needed to

¹³For all three models, predictive CEs are drawn from the Beta-Beta hierarchy using the model’s estimated hyperparameter posterior. The hyperprior predictive draws from the hyperprior directly, before seeing data.

discriminate between them. In other words, for lotteries with moderate probabilities the two functions are empirically interchangeable, and the CE overlays show that equivalence directly. The Power PWF, which distorts probabilities uniformly rather than producing an inverse-S shape, pulls the L_2 CE approximately \$1 lower, reflecting the structural restriction of the 1-parameter specification at $p = 0.5$.

The overall pattern suggests that the CRRA utility function is doing most of the work for these lottery designs. The three PWF specifications separate only at the symmetric lottery, where probability weighting is most consequential, and even there the differences are modest. These comparisons are available directly in the interpretable units of certainty equivalents, without requiring the researcher to compute the forward map post-estimation or to compare structural parameters across models.

The CE comparison suggests that the second PWF parameter contributes modestly for these data and lottery designs, with the models agreeing closely on L_1 and L_3 and differing by approximately \$1 on L_2 . Section 5.4 makes this comparison formal with a Bayes factor for the two flexible models, which the shared quantity-space prior renders directly comparable, and summarizes the corresponding evidence on the Power form.

5.2 *Subject-Level Cross-Model Agreement*

Since all three models share the CRRA utility function, there is reason to expect closer agreement at the individual level than if the utility functions differed. To examine this, I compute the cross-model divergence for each subject, defined as the average (across lotteries) of the maximum pairwise absolute difference in

median posterior CEs across the three models.

Figure 2 shows the subject with the smallest cross-model divergence. All three models produce nearly identical CE posteriors across all three lotteries. For this subject, the choice of PWF specification has no substantive consequence for predictive inference.

Figure 3 shows the subject with the largest cross-model divergence. This subject’s 80 binary choices are consistent with multiple risk preference profiles, and the PWF specification determines which profile the posterior favors. Since the utility function is the same across all three models, the divergence arises entirely from how each PWF resolves the ambiguity in the decision weights. The quantity-space prior is identical across models, so these differences reflect the structural restrictions of the functional forms rather than differences in prior beliefs.

These two cases illustrate a broader pattern. For well-identified subjects, the PWF specification has little effect on predictive inferences in quantity space. For weakly identified subjects, the functional form of probability weighting matters precisely because different PWFs resolve the same data ambiguity in different directions. The quantity-space framework makes both the agreement and the disagreement visible in dollars.

5.3 Sensitivity to α and κ

The preceding cross-model analysis holds the variance decomposition parameters (α, κ) fixed. To assess how sensitive the results are to these choices, I re-estimate Model 1 (CRRA + Prelec) at three additional configurations spanning a wide range of the (α, κ) space. For this exercise I introduce a diffuse parameter-space

Figure 2: Subject-level CE posteriors: most agreed subject (Subject 50)

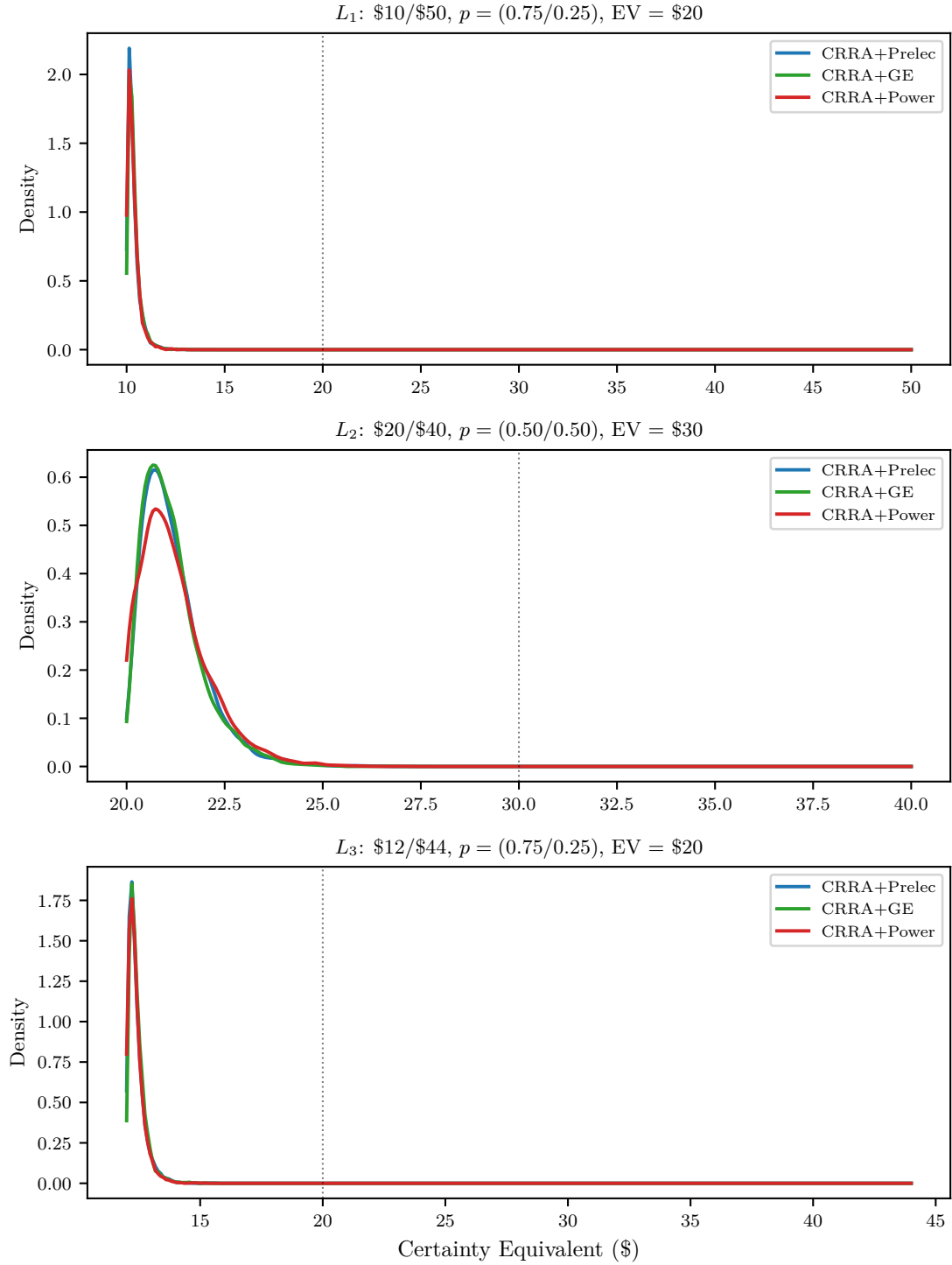
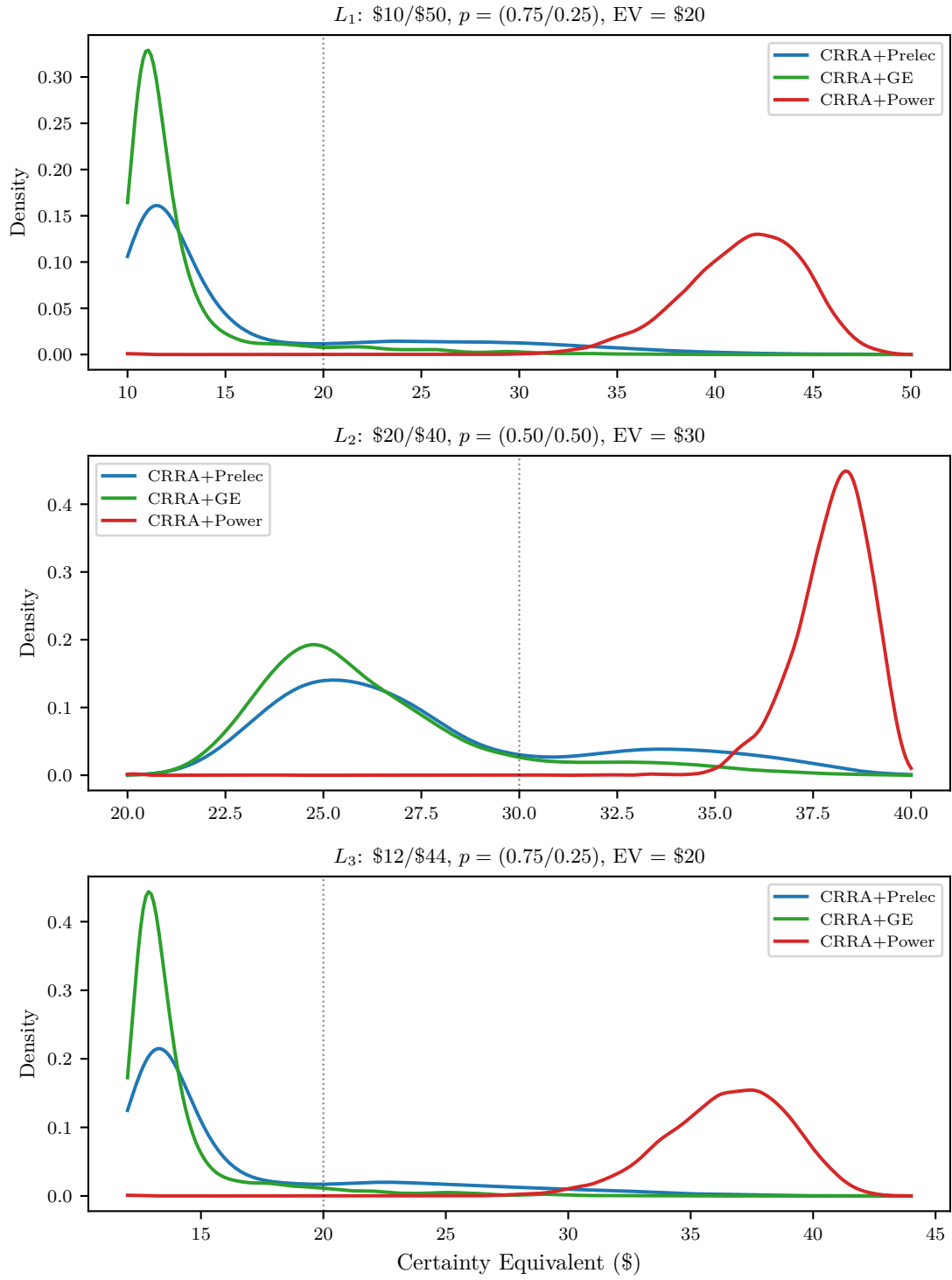


Figure 3: Subject-level CE posteriors: most divergent subject (Subject 55)



benchmark following Gao, Harrison and Tchernis (2022), which provides a fixed reference point against which to measure the influence of the informative prior at each configuration. Subjects are classified by the average absolute median CE difference between the informative and diffuse posteriors across the three reference lotteries: well-identified (less than \$2), moderately affected (\$2–\$10), and prior-dominated (more than \$10). Table 3 reports these classifications.

α	κ	Well-identified	Moderate	Prior-dominated	% dominated
0.2	0.25	80	13	18	16%
0.2	1.50	83	14	14	13%
0.8	0.25	76	14	21	19%
0.8	1.50	81	13	17	15%

Table 3: Subject classification across (α, κ) configurations for Model 1 (CRRA + Prelec) relative to a diffuse benchmark. Each row uses the same direct prior and data; only the hierarchical variance allocation changes.

The prior-dominated share ranges from 13% to 19% across the four configurations, confirming that the finding is structurally stable rather than an artifact of a particular (α, κ) choice. The direction of movement is consistent with the theory. Holding κ fixed, higher α (tighter population prior, less heterogeneity) increases the prior-dominated share from 16% to 19% at low κ and from 13% to 15% at high κ , because the narrower population distribution assigns lower density to extreme parameter values. Holding α fixed, higher κ (less confidence in the heterogeneity level) reduces the prior-dominated share from 16% to 13% at low α and from 19% to 15% at high α , because the looser variance hyperprior gives the data more room to pull weakly-identified subjects away from the prior mode. The moderate group is notably stable at 13–14 subjects across all four configurations, suggesting a structural middle band of subjects whose data

are informative enough to resist moderate prior perturbations. The full CDF difference plots and predictive CE overlays for each configuration are reported in Appendix D.

5.4 *Model Comparison via Bayes Factors*

A persistent concern with Bayes factors is their sensitivity to the prior specification, a point that Kass and Raftery (1995) discuss at length. In hierarchical models the concern is amplified, because the hyperprior governs the prior for every subject in the sample. When two candidate models carry separately specified parameter-space hyperpriors, a Bayes factor mixes evidence about structural form with differences in prior calibration. The three models estimated here inherit a single quantity-space prior over $(\text{CE}_1, \text{CE}_2, \text{CE}_3, P)$, so differences in marginal likelihood isolate the probability weighting specification. The comparison is therefore a test of functional form, not of prior calibration.

One qualification makes the shared-prior statement precise. The change-of-variables construction supports each subject’s prior on the model’s feasible set, the image of its forward map, which differs across models and is a 3-dimensional manifold inside the 4-dimensional quantity space for Model 3. The model as estimated is therefore the quantity-space hierarchy truncated to its feasible set. Truncation has two effects that the marginal likelihoods must account for. It renormalizes the conditional subject prior by the feasible mass $C_m(\boldsymbol{\gamma}) = \int p_{\vartheta}(\mathbf{g}_m(\boldsymbol{\theta}) \mid \boldsymbol{\gamma}) \text{vol } J_{\mathbf{g}_m}(\boldsymbol{\theta}) d\boldsymbol{\theta}$, and it reweights the hyperprior by $C_m(\boldsymbol{\gamma})^N$, with normalizer $Z_m = \mathbb{E} [C_m(\boldsymbol{\gamma})^N \mid \boldsymbol{\xi}]$ taken over the nominal hyperprior. At the posterior mean of the hyperparameters the feasible mass is 0.72 for Model 1 and 0.72 for Model 2, while for Model 3 the corresponding surface integral is 0.39.

Appendix E.2 describes how C_m and Z_m are estimated. For Models 1 and 2 the normalizer is estimated directly and the two models can be compared on fully normalized terms. For the over-determined Model 3 the normalizer can only be bounded, for reasons developed in Appendix E.2, so the formal comparison below is restricted to the two flexible models and the evidence on Model 3 is summarized afterward.

The object of comparison is the marginal likelihood of each model, which integrates the likelihood over every subject-level parameter and hyperparameter:

$$p(y \mid M_m) = \int p(y \mid \boldsymbol{\theta}, M_m) p(\boldsymbol{\theta} \mid \boldsymbol{\gamma}, M_m) p(\boldsymbol{\gamma} \mid M_m) d\boldsymbol{\theta} d\boldsymbol{\gamma} \quad (29)$$

For Models 1 and 2 this integral spans 452 dimensions, and for Model 3 it spans 341 (Table E.1). The log Bayes factor between models m and m' is the difference in log marginal likelihoods:

$$\log \text{BF}_{mm'} = \log p(y \mid M_m) - \log p(y \mid M_{m'}) \quad (30)$$

The Power PWF is nested within the Prelec family at $\phi = 1$, while the Prelec and Goldstein-Einhorn forms are non-nested. The marginal likelihood treats both cases identically, so a single set of estimates supports all three pairwise comparisons.

I estimate each marginal likelihood with a Rao-Blackwellized estimator that exploits the hierarchical structure. Conditional on the hyperparameters the subjects are independent, so the integral in Equation (29) factors into per-subject integrals over three or four parameters followed by a single 8-dimensional integral over the hyperparameters. The per-subject integrals are estimated by importance sampling with mixture proposals adapted to each subject's posterior. The hyperparameter

integral is then computed twice, by two routes that share no integration logic. The first is importance sampling from a Student- t distribution fitted to the hyperparameter posterior, and the second is the iterative bridge sampling estimator of Meng and Wong (1996). The whole procedure runs on 8 independent sets of importance draws, and the replicate spread provides the standard errors. As a correctness check, the same estimator recovers a brute-force benchmark on a small synthetic dataset to within 0.03 log units. Appendix E describes the procedure in detail.

Table 4 reports the log marginal likelihoods under both routes. The two routes agree to within 0.11 log units for every model, and the ranking of the models is the same under each.

	Importance sampling	Bridge sampling	$\log Z_m$
Model 1 (CRRA + Prelec)	-5218.2 (0.03)	-5218.3 (0.02)	-36.5 (0.10)
Model 2 (CRRA + GE)	-5234.3 (0.02)	-5234.4 (0.02)	-37.0 (0.34)

Table 4: Normalized log marginal likelihoods for the two flexible models, by integration route, with the feasible-mass normalizer $\log Z_m$. Replicate-based standard errors in parentheses; the $\log Z_m$ standard error is common to both routes.

The importance sampling estimates imply a log Bayes factor of 16.1 (± 0.4) for Prelec over Goldstein-Einhorn. On the scale of Kass and Raftery (1995) this is decisive evidence, and it answers the question the CE overlays could not. Section 5.1 found the two flexible forms empirically interchangeable at these lotteries, with nearly identical posterior predictive CEs, yet integrated over the full choice data the Prelec form is clearly favored. The comparison is exactly the situation the shared quantity-space prior is designed for, since the two non-nested specifications differ only in functional form and their feasible masses

nearly coincide, so neither prior calibration nor normalization plays any role in the verdict.

The evidence on Model 3 is summarized rather than tabulated, because the normalizer of the $k > p$ construction is available only as a bound. The Power PWF is nested in Prelec at $\phi = 1$, the cross-model CE analysis shows it departing from the flexible forms exactly where probability weighting is most consequential, and the bounded comparison in Appendix E.2 puts the log Bayes factor for Prelec over Power at no less than 107.4, an order of magnitude beyond the decisive threshold even at the bound. The data leave no doubt that the one-parameter restriction is rejected; what the $k > p$ case lacks is a sharp point estimate of the margin, a limitation taken up in Section 6.

Because the prior is fixed in quantity space, these numbers are interpretable as evidence about probability weighting alone. In other words, the Bayes factor answers in units of evidence the question the CE overlays raised in dollars.

6 Discussion

The main contribution of the framework is not to reduce the number of prior choices, but to reorganize them into objects that are easier to reason about. Instead of choosing hyperpriors directly on latent population parameters, the researcher first states beliefs about predictive quantities for a randomly selected individual, and then states how much of the resulting uncertainty should be attributed to population heterogeneity rather than uncertainty about the population mean. This separation is useful because researchers often have substantive intuition in experimental units even when they have little intuition about hyperprior parameters themselves. A further benefit, illustrated by the cross-model application,

is that the same quantity-space specification can serve as the prior for multiple structural models. The three specifications considered here differ in their probability weighting functions and even the number of free parameters, yet they share a single set of beliefs about certainty equivalents and choice probabilities. That sharing is also what made the Bayes factor comparison of Section 5.4 interpretable, since the marginal likelihoods differ only through the structural specification. The framework therefore extends the Bayesian workflow of Gabry et al. (2019) to the hierarchical setting by making the two distinct roles of a hyperprior explicit: anchoring the unconditional distribution of observables and allocating uncertainty across levels of the hierarchy.

In practice, the method complements the iterative prior-predictive workflow of Gabry et al. (2019) rather than replacing it. The researcher chooses a quantity space such as certainty equivalents, choice probabilities, or indifference amounts, and for each dimension provides a mean and variance reflecting beliefs about a single random individual, informed by published empirical estimates, pilot data, or expert judgment. The researcher then chooses α and κ to express beliefs about the degree of population heterogeneity and confidence in that specification. Values of $\alpha = 0.2$ and $\kappa = 0.25$, as used in the application, correspond to substantial heterogeneity with moderate confidence. The implied prior predictive distribution should then be examined and the inputs revised, exactly the loop that Gabry et al. (2019) make explicit. What the framework changes is where that iteration happens. Instead of iterating over $4p$ hyperprior parameters whose individual effects on observables are difficult to anticipate, the researcher iterates over direct prior beliefs about the k predictive quantities, with the variance decomposition handling the hierarchical layer automatically. The sensitivity

analysis in Appendix D indicates that the broad posterior patterns are stable across a range of (α, κ) values, even though the magnitude of prior influence moves in the expected direction. Estimation then proceeds with standard MCMC sampling, adding the Jacobian correction term to the log-posterior when quantity-space specification is used.

6.1 Limitations

The framework has several limitations. First, the variance decomposition is *moment-based*, matching the first two moments of the direct prior. If the researcher’s beliefs are better captured by higher moments (skewness, heavy tails), two-moment matching may be insufficient. The framework does generalize to higher moments via the law of total cumulance, but at the cost of additional researcher choices and numerical rather than closed-form solutions. An alternative is to match cumulative probabilities at distinct points of the direct-prior CDF, which would also permit distributions such as the Cauchy that lack the moments required by the current method.

Second, the current framework is built around independent marginal direct priors over the k predictive quantities. In the application, CE_1 , CE_2 , CE_3 , and P are treated as independent in the elicited prior even though they are deterministic functions of the same underlying parameters and are therefore necessarily dependent. This use of independent marginals is deliberate: in many applications, a researcher may be able to articulate marginal beliefs about several observables more easily than their full joint dependence. Those independent marginal beliefs do not imply an independent parameter prior. The Jacobian correction transports them through the shared structural map, so the induced prior on the parameter vector

is generally dependent and can exhibit nonzero covariance across parameters. What the elicitation step does not do is directly encode the researcher’s joint beliefs about the quantity vector itself. In particular, if two reference quantities are very similar, independent marginal priors may impose unintended constraints on the implied parameter prior. In practice, this risk can be mitigated by choosing sufficiently distinct reference lotteries and by checking the implied joint prior through prior predictive draws. Extending the framework to allow direct joint elicitation in quantity space is an important direction for future work.

Finally, quantity-space specification requires an injective mapping from parameters to predictive quantities. This is not a cosmetic condition; it constrains the design of reference quantities and implies that not all experimental designs are equally suitable for prior specification. Proving injectivity can also be technically demanding. The analytic argument in Appendix B covers all three models here because they share the scalar CRRA utility, but utility functions with more than one parameter, such as the Expo-Power form, replace that scalar argument with a higher-dimensional condition for which numerical verification of the Gram determinant is the practical route. Other quantities, such as choice probabilities, avoid some of these complications and can support flexible reference designs, at least for EUT. Appendix C shows that broad classes of lottery pairs yield injective maps when the predictive quantities are choice probabilities under EUT.

A related limitation concerns the model comparison of Section 5.4. The shared quantity-space prior removes cross-model differences in prior calibration from the comparison, but the resulting evidence is still conditional on the one shared specification. A different direct prior or a different (α, κ) allocation would

change the marginal likelihoods, and not necessarily by equal amounts, so the Prelec and Goldstein-Einhorn comparison should be read as evidence under the stated prior rather than as a prior-free verdict. The nested Power comparison is robust in the opposite sense, with the restriction $\phi = 1$ rejected by margins far larger than plausible prior perturbations, but it carries its own caveat. The feasible-mass tilt of the truncated $k > p$ construction concentrates at extreme heterogeneity values, its normalizer is bounded rather than point estimated, and a fully normalized comparison of over-determined specifications is a methodological problem the quantity-space framework inherits rather than solves.

6.2 Extensions

The application already demonstrates that a single quantity-space prior can serve models with different parameter dimensions. Model 3 maps 3 structural parameters to 4 predictive quantities through the Gram determinant, and the comparison of Section 5.4 ranks it against the two 4-parameter models, with the caveat that its normalizer is bounded rather than point estimated. The natural next step is to widen the comparison set. Models in the Cumulative Prospect Theory class, with separate utility and weighting functions over gains and losses, predict the same certainty equivalents and choice probabilities and could inherit the same quantity-space prior, despite having two to three times as many structural parameters.

Beyond model selection, the shared prior opens the door to model combination. Bayesian model averaging and stacking both require marginal likelihoods or predictive scores computed under comparable priors, and the quantity-space specification supplies exactly that comparability. A related direction is to carry

the same direct prior across datasets. The discounting experiment of Andersen, Harrison, Lau and Rutström (2008) elicits risk preferences over a different currency and stake range, and re-expressing the direct prior over that design’s reference lotteries would test how portable quantity-space beliefs are across experimental contexts. Finally, extending the elicitation step to joint beliefs over the quantity vector, rather than independent marginals, would address the limitation discussed above and sharpen all of these comparisons.

7 Conclusion

BHMs require hyperprior specifications that are difficult to construct from first principles. I have developed a two-part framework that reorganizes this problem into objects that are easier to reason about. First, quantity-space prior specification allows researchers to state beliefs about predictive quantities such as certainty equivalents and choice probabilities, with a Jacobian correction translating those beliefs exactly to parameter space. Second, a variance decomposition expresses how unconditional uncertainty is allocated across levels of the hierarchy, separating beliefs about population heterogeneity from beliefs about the location of the population mean.

The application to RDU estimation with three structurally different models illustrates the methodological value of this approach. The specifications differ in their probability weighting functions and in the number of free parameters, yet they share a single quantity-space prior over certainty equivalents and choice probabilities. The results show that informative hyperpriors matter most where the data are weak, while leaving well-identified subjects largely unchanged, and the shared prior supports a Bayes factor comparison in which differences

in marginal likelihood reflect structural form rather than differences in prior calibration.

The framework changes the object of prior specification. Instead of asking researchers to work directly with opaque hyperprior parameters, it asks them to reason about observables and about the structure of heterogeneity in the population. That shift makes hierarchical prior specification easier to interpret and diagnose, and it makes prior beliefs coherent across competing structural models that speak to the same behavioral quantities. This is the main advantage of the framework, and the reason quantity-space hyperprior specification may be useful well beyond the present application.

Code and Data Availability

The estimation code, data, and replication materials for this paper are available at <https://github.com/bmonroe3/bhm-prior-selection>.

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Appendix A: Hyperprior Derivation by Distributional Family

This appendix provides the complete derivation of hyperprior parameters from (α, κ) for three distributional families covering the most common support types in applied work: the Normal-Inverse-Gamma (support on $\mathbb{R}$), Gamma-Inverse-Gamma (support on $(0, \infty)$), and Beta-Beta (support on $[0, 1]$). Each subsection presents the hierarchy, the closed-form translation from researcher inputs to hyperprior parameters, the feasibility region in (α, κ) space, and graphical calibrations.

A.1 Normal-Inverse-Gamma (support on $\mathbb{R}$)

Consider the standard Normal-Inverse-Gamma hierarchical model:

$$\boldsymbol{\vartheta}_i \mid \mu, \sigma^2 \sim N(\mu, \sigma^2) \quad (\text{A.1})$$

$$\mu \mid \mu_0, \tau_0^2 \sim N(\mu_0, \tau_0^2) \quad (\text{A.2})$$

$$\sigma^2 \mid \alpha_0, \beta_0 \sim \text{IG}(\alpha_0, \beta_0) \quad (\text{A.3})$$

Here $\boldsymbol{\gamma}_m = \mu$ and $\boldsymbol{\gamma}_v = \sigma^2$, so the hyperprior parameters are $\boldsymbol{\xi} = (\mu_0, \tau_0^2, \alpha_0, \beta_0)$.

From (7), the mean hyperprior parameters follow immediately, since the Normal mean and variance are the distribution's parameters:

$$\mu_0 = m^* \quad (\text{A.4})$$

$$\tau_0^2 = \alpha v^* \quad (\text{A.5})$$

The conditional variance is the parameter σ^2 itself, so $\kappa = \text{SD}[\sigma^2]/\mathbb{E}[\sigma^2]$ maps directly to the Inverse-Gamma hyperprior. Using $\mathbb{E}[\sigma^2] = \beta_0/(\alpha_0 - 1)$ and

$\text{Var}[\sigma^2] = \beta_0^2/[(\alpha_0 - 1)^2(\alpha_0 - 2)]$, equations (10) yield:

$$\alpha_0 = 2 + \frac{1}{\kappa^2} \quad (\text{A.6})$$

$$\beta_0 = (1 - \alpha) v^* (\alpha_0 - 1) \quad (\text{A.7})$$

The shape parameter α_0 is determined solely by κ , with β_0 then following from α and v^* .

The NIG family admits the full region $\alpha \in (0, 1)$, $\kappa \in (0, \infty)$. The Inverse-Gamma requires $\alpha_0 > 2$ for the variance of σ^2 to exist, which holds for all $\kappa > 0$ since $\alpha_0 = 2 + 1/\kappa^2 > 2$ automatically. There are no joint constraints linking α and κ .

Figure A.1 displays the hierarchical variance allocation across the (α, κ) grid, setting $m^* = 0$ and $v^* = 1$. Each (α, κ) pair uniquely determines the hyperprior parameters $(\mu_0, \tau_0^2, \alpha_0, \beta_0)$ through the moment conditions (7)–(10). Panel (a) shows that the between-population SD depends only on α , as the horizontal color bands confirm: κ plays no role. This follows directly from the moment condition $\tau_0 = \sqrt{\alpha v^*}$. Panel (b) shows the within-population SD, $\mathbb{E}[\sigma]$: higher α allocates more variance to population-mean uncertainty, leaving less for heterogeneity, so the within-population SD decreases as α increases. Panel (c) reveals the role of κ . At low κ the within-population SD is tightly concentrated (the IQR is small), reflecting near-certainty about the level of heterogeneity. As κ grows the IQR increases, reflecting greater uncertainty about how heterogeneous the population is.

Figure A.2 quantifies the distributional discrepancy between the hierarchical predictive and the direct prior. Panel (a) shows the Kolmogorov–Smirnov distance

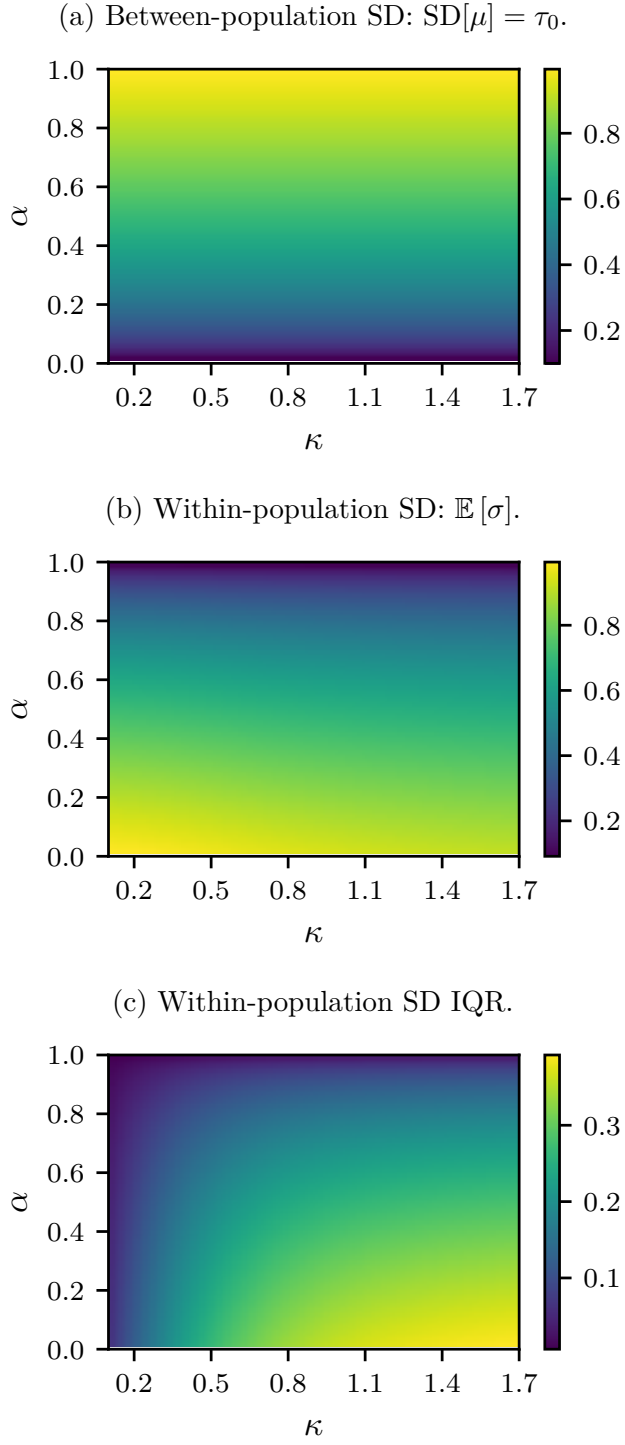


Figure A.1: Hierarchical variance allocation for the NIG model across (α, κ) , with $m^* = 0$ and $v^* = 1$.

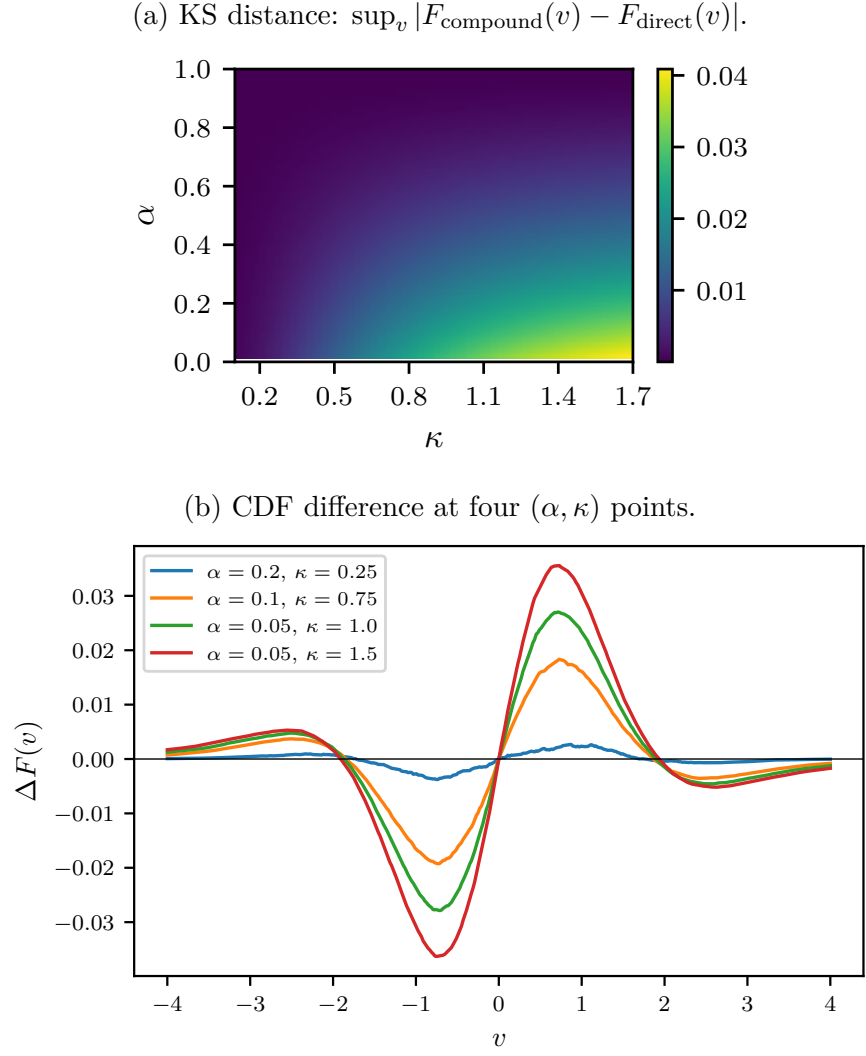


Figure A.2: Distributional approximation quality for the NIG hierarchy,
 $m^* = 0, v^* = 1$.

across the full (α, κ) grid. For moderate values such as those used in the application ($\alpha = 0.2, \kappa = 0.25$), the supremum CDF difference is approximately 0.3 percentage points. Even at the extreme corner ($\alpha \rightarrow 0, \kappa = 1.7$), it remains below 5 percentage points. Panel (b) reveals where the discrepancy arises: the compound distribution is flatter-peaked and heavier-tailed than the direct prior, consistent with the scale-mixture structure of the hierarchy. The CDF difference is negative in the shoulders ($|v| \lesssim 1.8\sigma^*$), where the compound places less mass, and positive in the tails ($|v| \gtrsim 1.8\sigma^*$), where it places more. All four curves share this qualitative shape; only the amplitude varies with α and κ .

The first two moments of the marginal prior on individual quantities are invariant across the grid by construction; only the hierarchical allocation changes. The specific numerical values are family-dependent, but these qualitative patterns generally hold for any distributional choice satisfying the moment conditions.

A.2 *Gamma-Inverse-Gamma (support on $(0, \infty)$)*

For positive-valued quantities such as income, prices, or time preferences, the Gamma-Inverse-Gamma hierarchy provides a natural model:

$$v_i \mid \mu, \sigma^2 \sim \text{Gamma}\left(\frac{\mu^2}{\sigma^2}, \frac{\mu}{\sigma^2}\right) \quad (\text{A.8})$$

$$\mu \mid a_\mu, b_\mu \sim \text{Gamma}(a_\mu, b_\mu) \quad (\text{A.9})$$

$$\sigma^2 \mid \alpha_0, \beta_0 \sim \text{IG}(\alpha_0, \beta_0) \quad (\text{A.10})$$

where the individual-level Gamma is parameterized by shape and rate so that $\mathbb{E}[v_i \mid \mu, \sigma^2] = \mu$ and $\text{Var}[v_i \mid \mu, \sigma^2] = \sigma^2$. The conditional variance is the parameter σ^2 itself, as in the NIG case.

The mean hyperprior uses $\mathbb{E}[\mu] = m^*$ and $\text{Var}[\mu] = \alpha v^*$, giving shape a_μ and

rate b_μ :

$$a_\mu = \frac{(m^*)^2}{\alpha v^*} \quad (\text{A.11})$$

$$b_\mu = \frac{m^*}{\alpha v^*} \quad (\text{A.12})$$

Because σ^2 is a parameter with its own independent hyperprior, the κ translation is identical to the NIG case:

$$\alpha_0 = 2 + \frac{1}{\kappa^2} \quad (\text{A.13})$$

$$\beta_0 = (1 - \alpha) v^* (\alpha_0 - 1) \quad (\text{A.14})$$

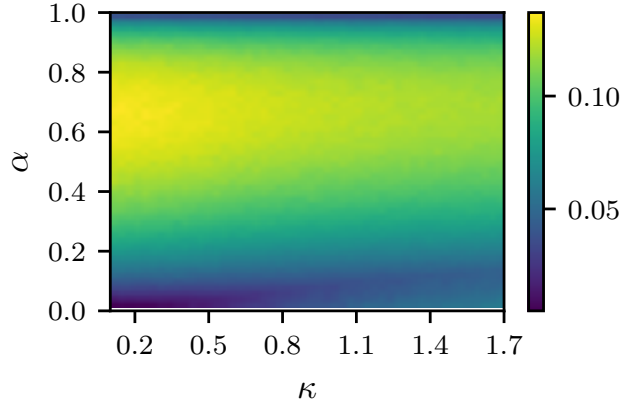
As with NIG, the Inverse-Gamma variance hyperprior admits all $\kappa > 0$. The Gamma mean hyperprior requires $a_\mu > 0$ and $b_\mu > 0$, which hold whenever $m^* > 0$ and $\alpha v^* > 0$. The feasibility region is therefore the full rectangle $\alpha \in (0, 1)$, $\kappa \in (0, \infty)$, the same as NIG.

The within-population statistics (within-population SD and its IQR) depend only on the Inverse-Gamma hyperprior with $\alpha_0 = 2 + 1/\kappa^2$ and $\beta_0 = (1 - \alpha) v^* (\alpha_0 - 1)$. These are identical to the NIG case, so the sensitivity heatmaps in Figure A.1 apply without modification: panel (a) gives the between-population SD, panel (b) the within-population SD, and panel (c) the within-population SD IQR for the Gamma-IG hierarchy as well. The between-population SD is $\sqrt{\alpha v^*}$ regardless of family, so these sensitivity surfaces are family-invariant.

The two families differ in the shape of the compound distribution: the NIG compound is symmetric with Student- t -like tails, while the Gamma-Inverse-Gamma compound is right-skewed.

Figure A.3 shows the distributional approximation quality for the Gamma-

(a) KS distance: $\sup_v |F_{\text{compound}}(v) - F_{\text{direct}}(v)|$.



(b) CDF difference at four (α, κ) points.

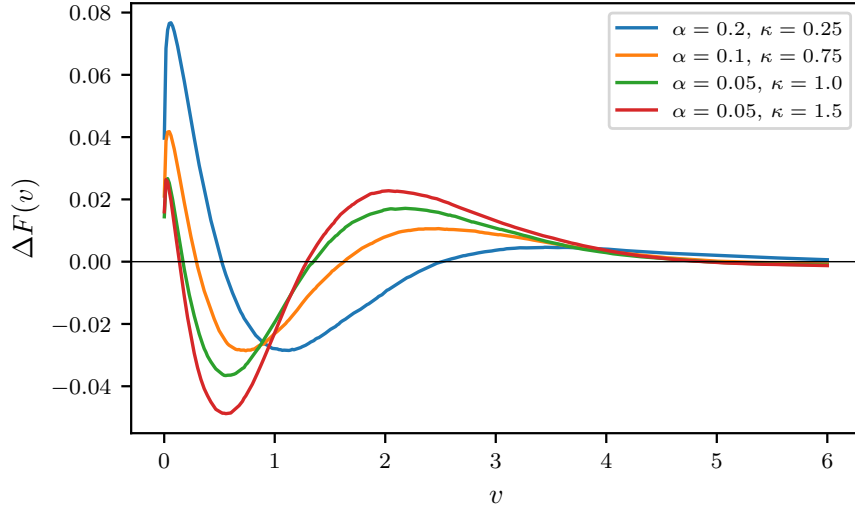


Figure A.3: Distributional approximation quality for the Gamma-IG hierarchy, $m^* = 1$, $v^* = 1$.

Inverse-Gamma case. Unlike the NIG, the Gamma-IG compound has a nonzero KS distance even at small (α, κ) : the minimum across the grid is approximately 0.005, reflecting an irreducible distributional mismatch between the Gamma compound and the direct Gamma prior that persists even as the hyperprior concentrates. The CDF difference curves are no longer antisymmetric, as the right-skewed compound distribution differs from the direct Gamma prior in a pattern that depends on the skewness of both distributions.

A.3 Beta-Beta (support on $[0, 1]$)

For bounded quantities normalized to $(0, 1)$, the Beta-Beta hierarchy uses a mean-dispersion parameterization. Let z_i^k denote the normalized value of quantity k for individual i :

$$z_i^k \mid \mu_k, \psi_k \sim \text{Beta}\left(\mu_k \frac{1-\psi_k}{\psi_k}, (1-\mu_k) \frac{1-\psi_k}{\psi_k}\right) \quad (\text{A.15})$$

$$\mu_k \sim \text{Beta}(a_\mu^k, b_\mu^k) \quad (\text{A.16})$$

$$\psi_k \sim \text{Beta}(a_\psi^k, b_\psi^k) \quad (\text{A.17})$$

where $\mu_k \in (0, 1)$ governs the population mean and $\psi_k \in (0, 1)$ governs dispersion. The conditional moments are $\mathbb{E}[z_i^k \mid \mu_k, \psi_k] = \mu_k$ and $\text{Var}[z_i^k \mid \mu_k, \psi_k] = \mu_k(1 - \mu_k)\psi_k$.

The mean-dispersion parameterization is convenient because it allows independent hyperpriors on μ_k and ψ_k . An alternative would be to parameterize the individual-level Beta in terms of its mean and variance directly, but $\text{Var}[z \mid \mu, \sigma^2] = \sigma^2$ would require $\sigma^2 < \mu(1 - \mu)$, creating a conditional domain that precludes independent hyperpriors. The normalization $\psi_k = \sigma_k^2 / [\mu_k(1 - \mu_k)]$ removes this dependence.

The law of total variance decomposes the direct prior variance:

$$v_k^* = \text{Var} [\mu_k] + \mathbb{E} [\mu_k(1 - \mu_k)\psi_k] \quad (\text{A.18})$$

$$\text{Var} [\mu_k] = \alpha_k v_k^* \quad (\text{A.19})$$

$$\mathbb{E} [\mu_k(1 - \mu_k)\psi_k] = (1 - \alpha_k) v_k^* \quad (\text{A.20})$$

Matching $\mathbb{E} [\mu_k] = m_k^*$ and $\text{Var} [\mu_k] = \alpha_k v_k^*$ by standard Beta moment-matching, the mean hyperprior parameters are:

$$a_\mu^k = m_k^* \left(\frac{m_k^*(1 - m_k^*)}{\alpha_k v_k^*} - 1 \right) \quad (\text{A.21})$$

$$b_\mu^k = (1 - m_k^*) \left(\frac{m_k^*(1 - m_k^*)}{\alpha_k v_k^*} - 1 \right) \quad (\text{A.22})$$

Using the independence of μ_k and ψ_k , the within-population term determines

$\mathbb{E} [\psi_k]$:

$$\mathbb{E} [\psi_k] = \frac{(1 - \alpha_k) v_k^*}{\mathbb{E} [\mu_k(1 - \mu_k)]} \quad (\text{A.23})$$

where $\mathbb{E} [\mu_k(1 - \mu_k)] = m_k^*(1 - m_k^*) - \alpha_k v_k^*$ is computed from the already-determined $\text{Beta}(a_\mu^k, b_\mu^k)$ hyperprior.

The parameter κ_k is the coefficient of variation of the conditional variance $\mu_k(1 - \mu_k)\psi_k$, as defined in Section 3.2. Because the conditional variance is a product of two independent random variables, its variance involves cross-moments. The complication is structural, not accidental, since Beta conditional variance depends on both the mean and the dispersion hyperparameters:

$$\text{Var} [\mu_k(1 - \mu_k)\psi_k] = \mathbb{E} [\mu_k^2(1 - \mu_k)^2] \text{Var} [\psi_k] + \text{Var} [\mu_k(1 - \mu_k)] \mathbb{E} [\psi_k]^2 \quad (\text{A.24})$$

Setting the left-hand side to $\kappa_k^2(1 - \alpha_k)^2(v_k^*)^2$ and solving for $\text{Var}[\psi_k]$:

$$\text{Var}[\psi_k] = \frac{\kappa_k^2(1 - \alpha_k)^2(v_k^*)^2 - \text{Var}[\mu_k(1 - \mu_k)] \mathbb{E}[\psi_k]^2}{\mathbb{E}[\mu_k^2(1 - \mu_k)^2]} \quad (\text{A.25})$$

The required moments are closed-form functions of (a_μ^k, b_μ^k) . For $\mu_k \sim \text{Beta}(a, b)$ with $S = a + b$:

$$\mathbb{E}[\mu_k(1 - \mu_k)] = \frac{ab}{S(S + 1)} \quad (\text{A.26})$$

$$\mathbb{E}[\mu_k^2(1 - \mu_k)^2] = \frac{a(a + 1)b(b + 1)}{S(S + 1)(S + 2)(S + 3)} \quad (\text{A.27})$$

$$\text{Var}[\mu_k(1 - \mu_k)] = \mathbb{E}[\mu_k^2(1 - \mu_k)^2] - \mathbb{E}[\mu_k(1 - \mu_k)]^2 \quad (\text{A.28})$$

Standard Beta moment-matching then gives the dispersion hyperprior parameters:

$$\nu_\psi^k = \frac{\mathbb{E}[\psi_k](1 - \mathbb{E}[\psi_k])}{\text{Var}[\psi_k]} - 1 \quad (\text{A.29})$$

$$a_\psi^k = \mathbb{E}[\psi_k] \nu_\psi^k \quad (\text{A.30})$$

$$b_\psi^k = (1 - \mathbb{E}[\psi_k]) \nu_\psi^k \quad (\text{A.31})$$

The Beta-Beta family imposes two constraints on (α, κ) . First, the mean hyperprior requires $\alpha v_k^* < m_k^*(1 - m_k^*)$, the Bernoulli bound on the variance of a $[0, 1]$ -valued random variable. This constrains α given the direct prior. Second, $\text{Var}[\psi_k] > 0$ in (A.25) requires:

$$\kappa > \frac{\sqrt{\text{Var}[\mu_k(1 - \mu_k)]}}{\mathbb{E}[\mu_k(1 - \mu_k)]} = \text{CV}[\mu_k(1 - \mu_k)] \quad (\text{A.32})$$

This lower bound on κ reflects the irreducible variation in conditional variance arising from μ_k alone: even with ψ_k fixed, the conditional variance $\mu_k(1 - \mu_k)\psi_k$ varies because μ_k is random. The bound increases with α (which increases $\text{Var}[\mu_k]$)

and is largest when m_k^* is near the boundary of $[0, 1]$. Additionally, $\text{Var}[\psi_k] < \mathbb{E}[\psi_k](1 - \mathbb{E}[\psi_k])$ is required for the Beta dispersion hyperprior to be proper, imposing an upper bound on κ for each (m_k^*, v_k^*, α_k) configuration. The feasibility region is therefore a proper subset of $(0, 1) \times (0, \infty)$, with both lower and upper bounds on κ that depend on α , m_k^* , and v_k^* . For typical prior configurations with m_k^* not too close to 0 or 1, the feasible range of κ is wide.

Figures A.4 and A.5 provide the graphical calibration for the Beta-Beta hierarchy with $m^* = 0.5$ and $v^* = 0.04$. The sensitivity heatmaps show qualitatively similar patterns to the NIG case, with the notable difference that regions of (α, κ) space where the Beta-Beta hyperprior is infeasible appear as missing values. The KS distances (panel (a) of Figure A.5) are substantially larger than the NIG case, reaching 0.20 at the extreme feasible corner compared to 0.04 for NIG. This reflects the additional distributional distortion introduced by the cross-moment coupling between μ_k and ψ_k : even after moment-matching, the Beta-Beta compound departs further from the direct prior than the NIG compound does.

A.4 Summary

The three families illustrate the range of behavior that arises when translating (α, κ) to hyperprior parameters. The NIG and Gamma-IG families share the simplifying property that the conditional variance is a single parameter (σ^2) with its own independent hyperprior, making κ map directly to the Inverse-Gamma shape parameter. Both admit the full (α, κ) region with no joint constraints. The Beta-Beta family, by contrast, has a conditional variance that depends on both hyperparameters, requiring a cross-moment correction and imposing both

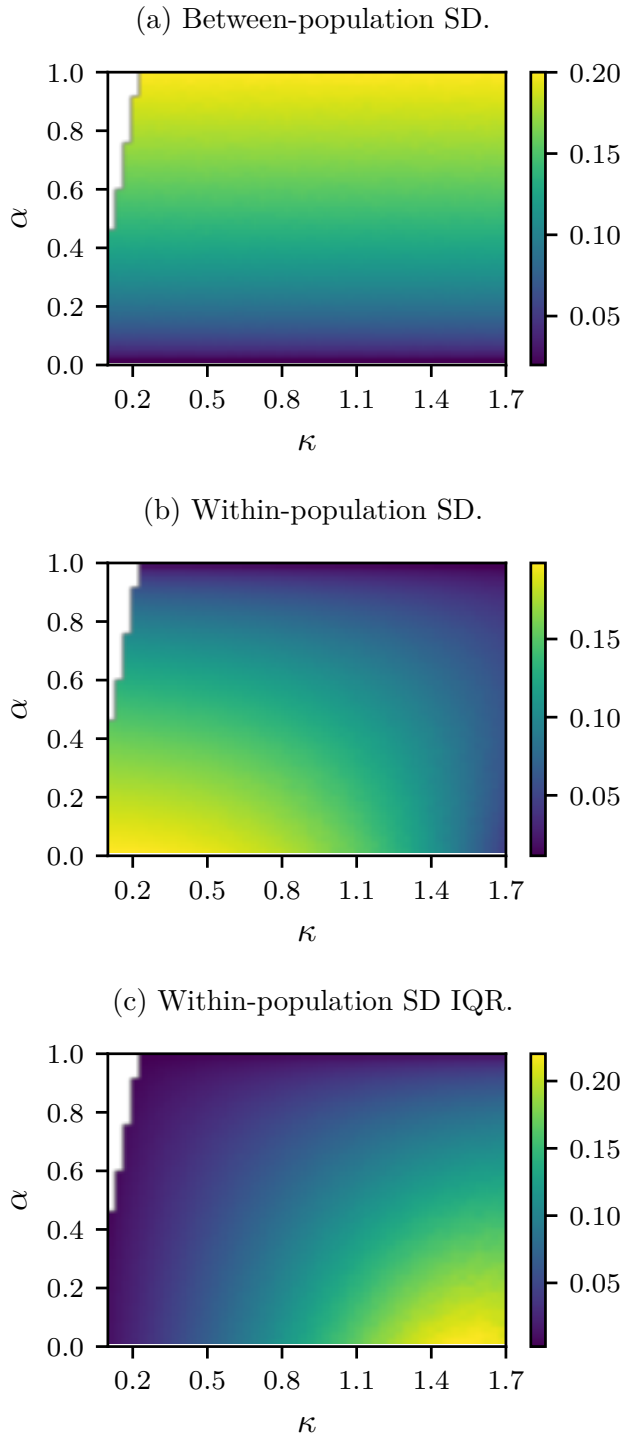
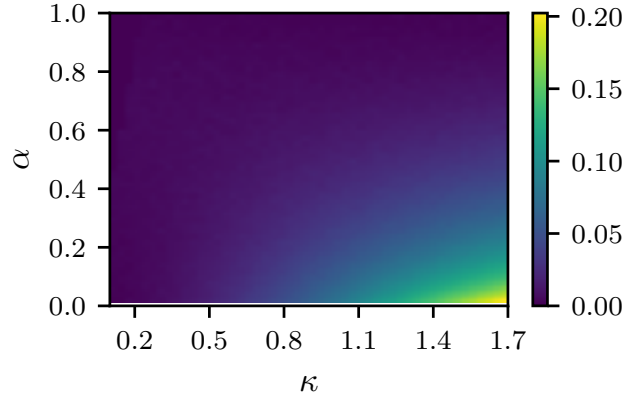


Figure A.4: Hierarchical variance allocation for the Beta-Beta model,
 $m^* = 0.5$, $v^* = 0.04$.

(a) KS distance: $\sup_z |F_{\text{compound}}(z) - F_{\text{direct}}(z)|$.



(b) CDF difference at four (α, κ) points.

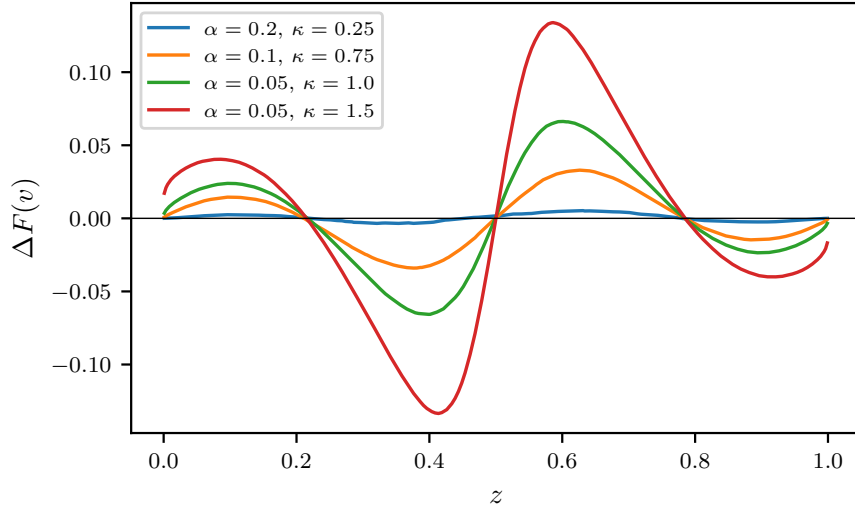


Figure A.5: Distributional approximation quality for the Beta-Beta hierarchy, $m^* = 0.5, v^* = 0.04$.

lower and upper bounds on κ , with the feasible region varying with α .

The distributional approximation quality follows a similar ordering. The NIG compound achieves the smallest KS distances (below 0.05 across the entire grid), the Gamma-IG is intermediate (0.005 to 0.14), and the Beta-Beta is largest (up to 0.20 at extreme feasible values). In all three cases, the moment-matching construction guarantees that the first two moments of the compound distribution match the direct prior exactly; only the higher moments differ, with the discrepancy concentrated in the tails and shoulders of the distribution. Thus, family choice governs two distinct objects: which (α, κ) combinations are feasible at all, and how closely the resulting compound predictive resembles the direct prior beyond its first two moments.

Appendix B: Injectivity Analysis for the Change-of-Variables Prior

For a scalar map $f: \mathbb{R} \rightarrow \mathbb{R}$, global injectivity is equivalent to strict monotonicity, which can be checked from the sign of the derivative. For maps $f: \mathbb{R}^n \rightarrow \mathbb{R}^n$ with $n > 1$, no such elementary criterion exists: a nonvanishing Jacobian determinant guarantees only local injectivity, and global conclusions require stronger conditions. The purpose of this appendix, however, is narrower than identification in the strongest everywhere sense: it is to provide sufficient conditions for the Jacobian-based change-of-variables prior used in the Bayesian analysis. For that application, injectivity is only needed on the support of the prior up to Lebesgue-null exceptional sets, since such sets do not affect an absolutely continuous prior density. The goal is therefore to show that the map from preference parameters to certainty equivalents (CEs) and choice probabilities is injective on a full-measure subset of the parameter space relevant for the transformed prior. I begin with two general propositions that are used to provide the proof.

Proposition B.1 (Composition). *If $g: A \rightarrow B$ and $f: B \rightarrow C$ are both injective, then the composite $f \circ g: A \rightarrow C$ is injective.*

Proof. Suppose $f(g(y)) = f(g(y'))$. Injectivity of f gives $g(y) = g(y')$. Injectivity of g gives $y = y'$. Q.E.D.

Proposition B.2 (Sequential decomposition). *Let $f: X \times Y \rightarrow Z$, and for each $x \in X$ define the conditional map $f_x: Y \rightarrow Z$ by $f_x(y) = f(x, y)$. Suppose:*

(i) *For each $x \in X$, the conditional map f_x is injective. Write $y = g(x, z)$ for the unique pre-image of each z in the range of f_x .*

(ii) *For each z in the range of f , the reduced map $x \mapsto f(x, g(x, z))$ is injective.*

Then f is injective.

Proof. Suppose $f(x, y) = f(x', y') = z$. Since $f_x(y) = z$, condition (i) gives $y = g(x, z)$; likewise $y' = g(x', z)$. Both x and x' satisfy $f(x, g(x, z)) = z$, so $x = x'$ by (ii). Then $y = g(x, z) = g(x', z) = y'$. Q.E.D.

The proposition reduces injectivity on a product space to two lower-dimensional problems: identify y conditional on x , then identify x from the residual.

B.1 Model, Conditions, Theorem, and Proof

The RDU model has three groups of parameters:

$$\boldsymbol{\theta} \in \Theta \subseteq \mathbb{R}^p \quad (\text{utility parameters}), \quad (\text{B.1})$$

$$\boldsymbol{\omega} \in \mathcal{W} \subseteq \mathbb{R}^q \quad (\text{PWF parameters}), \quad (\text{B.2})$$

$$\lambda \in (0, \infty) \quad (\text{noise parameter}). \quad (\text{B.3})$$

The utility function $u(x; \boldsymbol{\theta})$ is strictly increasing and differentiable in x for all $\boldsymbol{\theta} \in \Theta$. Let a and b denote the global minimum and maximum outcomes across

all K lotteries. Normalize utility on this common scale:

$$\tilde{u}(x; \boldsymbol{\theta}) = \frac{u(x; \boldsymbol{\theta}) - u(a; \boldsymbol{\theta})}{u(b; \boldsymbol{\theta}) - u(a; \boldsymbol{\theta})}, \quad (\text{B.4})$$

so that $\tilde{u}(a; \boldsymbol{\theta}) = 0$ and $\tilde{u}(b; \boldsymbol{\theta}) = 1$. Write $\mu_j(\boldsymbol{\theta}) = \tilde{u}(x_j; \boldsymbol{\theta}) \in [0, 1]$ for each outcome x_j appearing in any lottery.

The probability weighting function $w(p; \boldsymbol{\omega})$ is strictly increasing and differentiable in p for all $\boldsymbol{\omega} \in \mathcal{W}$, with $w(0; \boldsymbol{\omega}) = 0$ and $w(1; \boldsymbol{\omega}) = 1$. For each lottery L_k with ranked outcomes and probabilities, the decumulative probabilities and RDU decision weights are

$$S_i^{(k)} = \sum_{j \geq i} p_j^{(k)}, \quad (\text{B.5})$$

$$\pi_i^{(k)}(\boldsymbol{\omega}) = w(S_i^{(k)}; \boldsymbol{\omega}) - w(S_{i+1}^{(k)}; \boldsymbol{\omega}), \quad (\text{B.6})$$

with the convention $w(0) = 0$ and $w(1) = 1$.

The normalized RDU value of lottery k is

$$V_k(\boldsymbol{\theta}, \boldsymbol{\omega}) = \sum_{i=1}^{N_k} \pi_i^{(k)}(\boldsymbol{\omega}) \cdot \mu_i(\boldsymbol{\theta}), \quad (\text{B.7})$$

where the sum runs over the outcomes of lottery k , with $\mu_a = 0$ and $\mu_b = 1$.

The certainty equivalent c_k is the monetary amount satisfying $\tilde{u}(c_k; \boldsymbol{\theta}) = V_k$, i.e., $c_k = \tilde{u}^{-1}(V_k; \boldsymbol{\theta})$.

In a binary choice task, the subject observes lotteries A and B and chooses according to a noise model parameterized by λ : the choice probability $P = \Lambda((V_A - V_B)/(\lambda \Delta u))$ depends on $(\boldsymbol{\theta}, \boldsymbol{\omega})$ through the value difference and on λ through the noise scaling, where Λ is a strictly increasing link function and $\Delta u = u(\bar{x}) - u(\underline{x})$, where $\bar{x}$ and $\underline{x}$ are respectively the greatest and least outcomes

appearing with positive probability across the choice pair.

The goal is to show that the extended map

$$\tilde{F}: (\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda) \mapsto (c_1, \dots, c_K, P_{K+1}) \quad (\text{B.8})$$

is injective on a full-measure subset of $\Theta \times \mathcal{W} \times (0, \infty)$ sufficient for the Bayesian change of variables, where P_{K+1} is a choice probability from an additional task.

For the specific Prelec–CRRA design used in Section 4, three binary lotteries and one additional choice probability suffice. Two binary lotteries with distinct probabilities on the higher outcome produce two unique decision weights, which identify the two Prelec parameters. The verification below then uses a third binary lottery with shared probability $p_3 = p_1$ to identify the CRRA parameter. Because CRRA is invariant to multiplicative scaling of outcomes, the outcomes of L_3 cannot be a multiplicative rescaling of the outcomes of L_1 . A final non-degenerate choice task identifies the noise parameter away from the indifference locus. The remainder of this section establishes sufficient conditions for injectivity on a full-measure subset, which is the level needed for the transformed prior, first generally for any utility function and probability weighting function (Conditions A, B, and C below), followed by proofs that the proposed all-binary shared-probability design satisfies them (Sections B.2 and B.3) for the CRRA and 2-parameter Prelec functions.

B.1.1 Lottery design

We use $K = p + q$ lotteries in two blocks: q Block I lotteries $L_1, \dots, L_q$ ($k = 1, \dots, q$) that identify the PWF parameters, and p Block II lotteries ($k = q+1, \dots, p+q$) that identify the utility parameters. Each lottery satisfies (B.7);

the partition into blocks determines which parameters each group identifies.

The Block I lotteries must collectively evaluate w at q distinct probability points, enabling the factoring $\Phi_\mu = L \circ W$ (Section B.2.3). In the recommended *all-binary design*, each Block I lottery is binary (two-outcome) and evaluates w at exactly one probability point; with $q = 2$ binary lotteries having distinct probabilities $p_1 \neq p_2$, the L -matrix is diagonal and trivially invertible. Block II provides additional lotteries whose CEs depend on the utility parameters through a channel not already captured by Block I, identifying utility from the residual equations. A single additional binary choice task identifies the noise parameter λ (Condition C).

B.1.2 Condition A: PWF identification from Block I

When $\boldsymbol{\theta}$ is fixed, the normalized utilities $\mu_j(\boldsymbol{\theta})$ are known constants, and the q Block I instances of (B.7) become q equations in q unknowns $\boldsymbol{\omega}$. Condition A requires that these equations determine $\boldsymbol{\omega}$ uniquely, which is condition (i) of Proposition B.2.

Condition B.3 (A). For each fixed $\boldsymbol{\theta} \in \Theta$, the map $\Phi_\mu: \mathcal{W} \rightarrow \mathbb{R}^q$ defined by

$$[\Phi_\mu(\boldsymbol{\omega})]_k = \sum_i \pi_i^{(k)}(\boldsymbol{\omega}) \cdot \mu_i, \quad k = 1, \dots, q \quad (\text{B.9})$$

is injective.

When Condition A holds, each $\boldsymbol{\theta}$ determines a unique $\boldsymbol{\omega}_A(\boldsymbol{\theta}; \mathbf{c})$ solving the Block I system. The condition depends only on the PWF family and the Block I lottery structure, not on the observed CEs or the utility family.

B.1.3 Condition B: Utility identification from Block II

Substituting $\boldsymbol{\omega} = \boldsymbol{\omega}_A(\boldsymbol{\theta}; \mathbf{c})$ from Condition A into the p Block II instances of (B.7) eliminates the PWF parameters, yielding p equations in $\boldsymbol{\theta}$ alone. Condition B requires these residual equations to determine $\boldsymbol{\theta}$ uniquely, which is condition (ii) of Proposition B.2.

Condition B.4 (B). For any $\mathbf{c}$ in the range of F , the reduced map $H_{\mathbf{c}}: \Theta \rightarrow \mathbb{R}^p$ defined by

$$[H_{\mathbf{c}}(\boldsymbol{\theta})]_k = \sum_i \pi_i^{(k)}(\boldsymbol{\omega}_A(\boldsymbol{\theta}; \mathbf{c})) \cdot \mu_i(\boldsymbol{\theta}) - \tilde{u}(c_k; \boldsymbol{\theta}), \quad k = q + 1, \dots, p + q \quad (\text{B.10})$$

is injective.

Unlike Condition A, Condition B depends on both the utility and PWF families and on the observed CEs. When $\boldsymbol{\theta}$ changes, three things move simultaneously: the normalized utilities $\mu_i(\boldsymbol{\theta})$, the implied PWF $\boldsymbol{\omega}_A(\boldsymbol{\theta}; \mathbf{c})$, and the left-hand side $\tilde{u}(c_k; \boldsymbol{\theta})$. Condition B requires that these effects cannot cancel across all p Block II equations.

Condition A implies that W is injective on the Block I probability points (by contrapositive: identical decision weights produce identical Block I CEs). One may therefore reparameterize, replacing $\boldsymbol{\omega}$ with the decision weights. This is most effective when Block II probabilities coincide with Block I probabilities, so that no inversion to PWF parameters is required. Section B.2 establishes Condition A for the Prelec family, and Section B.3 exploits the resulting (r, w_1, w_2) reparameterization with a shared-probability design ($p_3 = p_1$).

Verifying Condition B requires the implicit derivative $d\boldsymbol{\omega}_A/d\boldsymbol{\theta}$, obtained by

differentiating the Block I system:

$$\frac{d\boldsymbol{\omega}_A}{d\boldsymbol{\theta}} = -J_A^{-1} \frac{\partial \Phi_\mu}{\partial \boldsymbol{\theta}}, \quad (\text{B.11})$$

where $J_A = \partial \Phi_\mu / \partial \boldsymbol{\omega}$ (invertible by Condition A).

B.1.4 Condition C: Noise identification from a choice probability

Once $(\boldsymbol{\theta}, \boldsymbol{\omega})$ are identified from the CEs, the value difference $\Delta V = V_A - V_B$ for any additional choice task is determined. Condition C requires that the choice probability $P = \Lambda(\Delta V / (\lambda \Delta u))$ identifies λ .

Condition B.5 (C). For the identified $(\boldsymbol{\theta}, \boldsymbol{\omega})$, the choice task has $\Delta V \neq 0$.

Proposition B.6 (Noise identification). *If Λ is strictly increasing and $\Delta V \neq 0$, then $P = \Lambda(\Delta V / (\lambda \Delta u))$ is strictly monotone in λ , and Condition C holds.*

Proof. The argument $\Delta V / (\lambda \Delta u)$ is strictly monotone in λ when $\Delta V \neq 0$; composing with the strictly increasing Λ preserves strict monotonicity. Q.E.D.

Condition C is pointwise, but the main text's change-of-variables prior $p(\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda) = p_Y(\tilde{F}(\cdot)) |\det J_{\tilde{F}}|$ requires injectivity to hold almost everywhere. Define the indifference locus $\mathcal{J} = \{(\boldsymbol{\theta}, \boldsymbol{\omega}) : \Delta V(\boldsymbol{\theta}, \boldsymbol{\omega}) = 0\}$. For the prior density to be well defined, $\mathcal{J}$ must have Lebesgue measure zero. Degenerate designs, such as $A = B$ or pairs whose RDU values coincide for all $(\boldsymbol{\theta}, \boldsymbol{\omega})$, give $\mathcal{J} = \Theta \times \mathcal{W}$ and must be excluded. For non-degenerate designs where ΔV is analytic and not identically zero, $\mathcal{J}$ is a lower-dimensional analytic variety (hence has Lebesgue measure zero), and Condition C holds on the full-measure complement. For standard analytic utility and PWF families, it is usually enough to choose a genuinely

distinct pair of lotteries for which ΔV is not identically zero as a function of $(\boldsymbol{\theta}, \boldsymbol{\omega})$, but this is a property of the specific design and should be checked rather than assumed.

B.1.5 Main theorem

Theorem B.7 (Injectivity sufficient for the Bayesian change of variables). *Suppose Conditions A and B hold on $\Theta \times \mathcal{W}$, and let*

$$\mathcal{J} = \{ (\boldsymbol{\theta}, \boldsymbol{\omega}) : \Delta V(\boldsymbol{\theta}, \boldsymbol{\omega}) = 0 \}.$$

If $\mathcal{J}$ has Lebesgue measure zero, then the extended map $\tilde{F}: (\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda) \mapsto (c_1, \dots, c_{p+q}, P_{K+1})$ is injective on the full-measure subset

$$((\Theta \times \mathcal{W}) \setminus \mathcal{J}) \times (0, \infty).$$

Consequently, for any prior on $(\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda)$ that is absolutely continuous with respect to Lebesgue measure, the transformed prior

$$p(\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda) = p_Y(\tilde{F}(\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda)) |\det J_{\tilde{F}}(\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda)|$$

is well defined almost everywhere.

Proof. The CEs do not depend on λ , so the map has block-triangular structure: apply Proposition B.2 with $X = (\boldsymbol{\theta}, \boldsymbol{\omega})$, $Y = \lambda$, and $Z = (c_1, \dots, c_{p+q}, P_{K+1})$.

Suppose

$$(\boldsymbol{\theta}, \boldsymbol{\omega}), (\boldsymbol{\theta}', \boldsymbol{\omega}') \notin \mathcal{J} \quad \text{and} \quad \tilde{F}(\boldsymbol{\theta}, \boldsymbol{\omega}, \lambda) = \tilde{F}(\boldsymbol{\theta}', \boldsymbol{\omega}', \lambda').$$

By Condition A, fixing $\boldsymbol{\theta}$ determines $\boldsymbol{\omega} = \boldsymbol{\omega}_A(\boldsymbol{\theta}; \mathbf{c})$ uniquely, and likewise $\boldsymbol{\omega}' = \boldsymbol{\omega}_A(\boldsymbol{\theta}'; \mathbf{c})$. Substituting into the Block II equations yields p equations in

θ alone; by Condition B, $\theta = \theta'$, and then $\omega = \omega'$. With (θ, ω) identified, ΔV is determined. Because $(\theta, \omega) \notin \mathcal{J}$, we have $\Delta V \neq 0$, so Proposition B.6 gives $\lambda = \lambda'$. Thus $\tilde{F}$ is injective on $((\Theta \times \mathcal{W}) \setminus \mathcal{J}) \times (0, \infty)$. The final statement follows because an absolutely continuous prior assigns probability zero to $\mathcal{J} \times (0, \infty)$, so null exceptional sets do not affect the induced prior measure. Q.E.D.

B.2 Verifying Condition A: Two-Point Injectivity and the $L \circ W$ Factoring

Conditions A and B require verification for specific utility and PWF families; Condition C requires a non-degenerate choice task ($\Delta V \neq 0$). This section and the next verify all three conditions for Prelec–CRRA.

The proof strategy is sequential. Condition A identifies the Prelec probability-weighting parameters from two probability points, reducing the Block I system to a diagonal problem. Condition B is the substantive step: once the decision weights are pinned down, it remains to show that the residual CE equation identifies the CRRA parameter through a monotonicity argument. That is done first by deriving an explicit identity for the derivative of the third CE, and then by proving monotonicity of the kernel appearing in that identity. Condition C then identifies the noise parameter from one additional non-degenerate choice task.

For $q = 2$, the map Φ_μ factors as $L \circ W$, where W is the two-point evaluation map and L is a 2×2 matrix determined by the lottery structure. When $\det(L) \neq 0$, Condition A reduces to injectivity of W .

B.2.1 Two-point injectivity

For the Prelec PWF $w(p; \phi, \eta) = \exp(-\eta t^\phi)$ with $t = -\ln p > 0$, define the *two-point evaluation map* $W: (\phi, \eta) \mapsto (w(p_1), w(p_2))$ for distinct $p_1 \neq p_2 \in (0, 1)$, and write $t_i = -\ln p_i$, $w_i = w(p_i)$.

Lemma B.8 (Prelec two-point injectivity). *The two-point evaluation map W is injective on $\mathbb{R}_{>0}^2$ for any $p_1 \neq p_2 \in (0, 1)$.*

Proof. Write $t_i = -\ln p_i > 0$. Suppose $W(\phi_1, \eta_1) = W(\phi_2, \eta_2)$. Taking logarithms of both components:

$$\eta_1 t_1^{\phi_1} = \eta_2 t_1^{\phi_2}, \tag{B.12}$$

$$\eta_1 t_2^{\phi_1} = \eta_2 t_2^{\phi_2}. \tag{B.13}$$

Dividing (B.12) by (B.13):

$$\left(\frac{t_1}{t_2}\right)^{\phi_1} = \left(\frac{t_1}{t_2}\right)^{\phi_2}. \tag{B.14}$$

Since $p_1 \neq p_2$, the values $t_1 \neq t_2$, so $t_1/t_2 \neq 1$. The function a^x is injective in x for any $a > 0$ with $a \neq 1$, so (B.14) implies $\phi_1 = \phi_2$. Substituting back into (B.12): $\eta_1 = \eta_2$, since $t_1^{\phi_1} > 0$. Q.E.D.

The proof uses only the *ratio property*: a logarithmic transformation separates the parameters multiplicatively, and the resulting basis functions form a ratio differing from 1 at distinct probability points. For the Goldstein–Einhorn family the analogous transformation is the logit, and the argument is recorded explicitly as Lemma B.9 below.

B.2.2 W is a bijection: the (r, w_1, w_2) reparameterization

Injectivity of W alone enables the change of variables from (ϕ, η) to (w_1, w_2) : one works in the image $W(\mathbb{R}_{>0}^2) \subseteq (0, 1)^2$, which may be a proper subset, and every statement about (r, w_1, w_2) applies to the parameter values actually attained by the PWF family. The simplification from working in decision-weight space is substantial enough to justify this reparameterization even when the image $W(\mathbb{R}_{>0}^2)$ must be characterized explicitly. When W is bijective, the image equals the full codomain and no such characterization is needed.

For the Prelec family, W is in fact *bijective* onto the order-compatible region

$$W(\mathbb{R}_{>0}^2) = \begin{cases} \{(w_1, w_2) : 0 < w_1 < w_2 < 1\}, & \text{if } p_1 < p_2, \\ \{(w_1, w_2) : 0 < w_2 < w_1 < 1\}, & \text{if } p_1 > p_2. \end{cases} \quad (\text{B.15})$$

The proof of Lemma B.8 provides the explicit inverse. Given observed values $w_1 = w(p_1)$ and $w_2 = w(p_2)$ with $t_i = -\ln p_i$:

$$\begin{aligned} \phi &= \frac{\ln(\ln w_1 / \ln w_2)}{\ln(t_1 / t_2)}, \\ \eta &= \frac{-\ln w_1}{t_1^\phi}. \end{aligned} \quad (\text{B.16})$$

Both expressions are well defined whenever $w_1, w_2 \in (0, 1)$ satisfy the ordering induced by p_1 and p_2 . The condition $p_1 \neq p_2$ ensures $t_1 \neq t_2$ so the denominator is nonzero, and $w_i \in (0, 1)$ ensures $-\ln w_i > 0$. If $p_1 < p_2$, then $t_1 > t_2$, so $\ln(t_1/t_2) > 0$ and positivity of ϕ requires $\ln(\ln w_1 / \ln w_2) > 0$, equivalently $w_1 < w_2$. The case $p_1 > p_2$ is symmetric, yielding $w_1 > w_2$. The Goldstein–Einhorn family admits the same bijectivity by a direct argument.

Lemma B.9 (Goldstein–Einhorn two-point injectivity and bijectivity). *For the*

Goldstein–Einhorn PWF $w(p; \gamma, \delta) = \delta p^\gamma / (\delta p^\gamma + (1 - p)^\gamma)$ with $\gamma, \delta > 0$, the two-point evaluation map $W_{\text{GE}}: (\gamma, \delta) \mapsto (w(p_1), w(p_2))$ is injective on $\mathbb{R}_{>0}^2$ for any $p_1 \neq p_2 \in (0, 1)$, and is a bijection onto the order-compatible region (B.15).

Proof. Write $\text{logit}(x) = \ln(x/(1 - x))$ and $\ell_i = \text{logit}(p_i)$. From the definition, $w/(1 - w) = \delta (p/(1 - p))^\gamma$, so

$$\text{logit}(w(p_i)) = \ln \delta + \gamma \ell_i, \quad i = 1, 2. \quad (\text{B.17})$$

Because the logit is strictly increasing, $p_1 \neq p_2$ gives $\ell_1 \neq \ell_2$, so these two equations form an invertible linear system in $(\ln \delta, \gamma)$ with explicit solution $\gamma = [\text{logit}(w_1) - \text{logit}(w_2)]/(\ell_1 - \ell_2)$ and $\ln \delta = \text{logit}(w_1) - \gamma \ell_1$. Injectivity follows. The constraint $\gamma > 0$ holds exactly when w_1 and w_2 share the ordering of p_1 and p_2 , and $\delta > 0$ is automatic, so every pair in the order-compatible region is attained. Q.E.D.

Because of this bijection, we conduct all subsequent analysis in (r, w_1, w_2) space subject only to the ordering restriction implied by p_1 and p_2 ; the Prelec parameters do not appear again. This ordering restriction is automatic for the Prelec family on $\mathbb{R}_{>0}^2$, so it does not impose any additional support restriction beyond remaining in the image $W(\mathbb{R}_{>0}^2)$.

For the all-binary design, each Block I lottery L_k is binary with probability p_k on the better outcome. When $p_1 \neq p_2$, the L -matrix in the $L \circ W$ factoring is $L = \text{diag}(\Delta\mu_1, \Delta\mu_2)$ with $\det(L) = \Delta\mu_1 \Delta\mu_2 > 0$ (Section B.2.3 below), and Condition A holds unconditionally.

Proposition B.10 (All-binary Condition A). *For the all-binary Block I design with $p_1 \neq p_2$, Condition A holds at all $(\phi, \eta) \in \mathbb{R}_{>0}^2$ and all utility parameters,*

with no restrictions on outcomes.

Proof. Each binary lottery evaluates w at one probability, giving $L = \text{diag}(\Delta\mu_1, \Delta\mu_2)$ with $\det(L) > 0$. By Lemma B.8 and the invertibility of L , Condition A holds.

Q.E.D.

B.2.3 The $L \circ W$ factoring

For a binary lottery L_k with outcomes $(\underline{x}_k, \bar{x}_k)$ and probability p_k on the better outcome, the normalized RDU value (B.7) is

$$\begin{aligned} V_k &= \mu_{\underline{x}_k} + w(p_k) \Delta\mu_k, \\ \Delta\mu_k &:= \mu_{\bar{x}_k} - \mu_{\underline{x}_k} > 0. \end{aligned} \tag{B.18}$$

With two binary lotteries having $p_1 \neq p_2$:

$$\begin{pmatrix} V_1 - \mu_{\underline{x}_1} \\ V_2 - \mu_{\underline{x}_2} \end{pmatrix} = \underbrace{\begin{pmatrix} \Delta\mu_1 & 0 \\ 0 & \Delta\mu_2 \end{pmatrix}}_L \underbrace{\begin{pmatrix} w(p_1) \\ w(p_2) \end{pmatrix}}_{W(\phi, \eta)}, \tag{B.19}$$

with $\det(L) = \Delta\mu_1 \Delta\mu_2 > 0$. For fixed $\boldsymbol{\theta}$, Condition A reduces to injectivity of W , which holds by Lemma B.8.

B.3 Application: Prelec-CRRA in (r, w_1, w_2) Space

For the concrete case in the main text: $\boldsymbol{\theta} = r$ (CRRA coefficient, $p = 1$) and $\boldsymbol{\omega} = (\phi, \eta)$ (Prelec parameters, $q = 2$), with utility $u(x; r) = x^{1-r}/(1-r)$ and PWF $w(p; \phi, \eta) = \exp(-\eta(-\ln p)^\phi)$. By the W -bijectivity of Section B.2.2, the model is reparameterized as $F: (r, w_1, w_2) \mapsto (c_1, c_2, c_3)$. All analysis in this section operates in (r, w_1, w_2) space.

B.3.1 The CE map

For a binary lottery $L_k = (\underline{x}_k, \bar{x}_k)$ with decision weight w_k on the better outcome, write $s = 1 - r$. The CRRA certainty equivalent is a power mean:

$$c_k^s = w_k \bar{x}_k^s + (1 - w_k) \underline{x}_k^s, \quad (\text{B.20})$$

equivalently $c_k = [w_k \bar{x}_k^s + (1 - w_k) \underline{x}_k^s]^{1/s}$. Since the right-hand side is strictly increasing in w_k , Condition A is immediate in (r, w_1, w_2) space: given r , each c_k is uniquely determined by w_k .

B.3.2 Condition B: CRRA identification via shared probability

With $p = 1$, Condition B reduces to the scalar condition $dc_3/dr|_{c_1} \neq 0$. The shared-probability design $p_3 = p_1$ eliminates the PWF parameters entirely: both L_1 and L_3 share the decision weight $w_3 = w_1$. The proof has three steps. First, show that $b = 0$ is fatal because then L_3 is only a rescaling of L_1 . Second, normalize out the scale factor m and derive a usable identity for dc_3/ds . Third, show that the kernel in that identity is strictly decreasing, so any nonzero shift b changes the outcome ratio and forces $dc_3/ds \neq 0$.

The first step is to establish the necessary condition and normalize away the irrelevant scale factor.

Parameterize L_3 's outcomes relative to L_1 's:

$$\begin{aligned} \bar{x}_3 &= m \bar{x}_1 + b, \\ \underline{x}_3 &= m \underline{x}_1 + b. \end{aligned} \quad (\text{B.21})$$

where $m = (\bar{x}_3 - \underline{x}_3)/(\bar{x}_1 - \underline{x}_1) > 0$ is the spread ratio and $b = \underline{x}_3 - m \underline{x}_1$ is the

shift. Substituting into (B.20) with $w_3 = w_1$:

$$c_1^s = w_1 \bar{x}_1^s + (1 - w_1) \underline{x}_1^s, \quad (\text{B.22})$$

$$c_3^s = w_1 (m \bar{x}_1 + b)^s + (1 - w_1) (m \underline{x}_1 + b)^s. \quad (\text{B.23})$$

Solving (B.22) for w_1 in terms of s :

$$w_1(s) = \frac{c_1^s - \underline{x}_1^s}{\bar{x}_1^s - \underline{x}_1^s}. \quad (\text{B.24})$$

Substituting into (B.23) makes c_3 a function of s alone:

$$c_3^s = w_1(s) [(m \bar{x}_1 + b)^s - (m \underline{x}_1 + b)^s] + (m \underline{x}_1 + b)^s. \quad (\text{B.25})$$

Setting $b = 0$ reveals a necessary condition. In that case L_3 is just the scaled lottery mL_1 , so

$$c_3^s = m^s [w_1(s) \bar{x}_1^s + (1 - w_1(s)) \underline{x}_1^s] = m^s c_1^s, \quad (\text{B.26})$$

hence $c_3 = m c_1$. When $b = 0$, the CE $c_3 = m c_1$ is independent of s , so L_3 provides no information about r . This reflects the well-known scale invariance of CRRA utility. The remainder of this section shows that $b \neq 0$ is also *sufficient* for identification.

CRRA homogeneity $\text{CE}(mL) = m \cdot \text{CE}(L)$ implies that the scale factor m does not affect monotonicity of c_3 in r , so we may normalize by dividing L_3 's outcomes by m . Writing $e = b/m$, the condition $b \neq 0$ is equivalent to $e \neq 0$. Define the shifted outcomes of this normalized lottery as $\underline{y}, \bar{y}$ to distinguish them

from the original L_3 outcomes $\underline{x}_3, \bar{x}_3$, which still satisfy $\underline{x}_3 = m \underline{y}$ and $\bar{x}_3 = m \bar{y}$:

$$\underline{y} = \underline{x}_1 + e,$$

$$\bar{y} = \bar{x}_1 + e.$$

Then the CE equations become

$$c_1^s = w_1 \bar{x}_1^s + (1 - w_1) \underline{x}_1^s, \quad (\text{B.27})$$

$$\tilde{c}_3^s = w_1 \bar{y}^s + (1 - w_1) \underline{y}^s, \quad (\text{B.28})$$

where $\tilde{c}_3$ is the CE of the normalized lottery (with $c_3 = m \tilde{c}_3$); write c_3 for $\tilde{c}_3$ hereafter. Because all CRRA outcomes must remain positive, $0 < \underline{y} < \bar{y}$.

The second step is to derive a usable formula for dc_3/ds .

The next lemma isolates the algebraic reduction. Its role is to rewrite dc_3/ds so that the sign question depends only on a simple prefactor and a one-dimensional kernel.

Lemma B.11 (Condition B derivative identity). *Under the normalized shared-probability representation above,*

$$\frac{dc_3}{ds} = \frac{\bar{y}^s - \underline{y}^s}{s^2 c_3^{s-1}} \left[H_{w_1(s)}(\rho_3) - H_{w_1(s)}(\rho_1) \right], \quad (\text{B.29})$$

where

$$\begin{aligned} \rho_1 &= \left(\frac{\underline{x}_1}{\bar{x}_1} \right)^s, \\ \rho_3 &= \left(\frac{\underline{y}}{\bar{y}} \right)^s, \\ H_w(\rho) &= \frac{(1-w)\rho \ln \rho - [w + (1-w)\rho] \ln[w + (1-w)\rho]}{1 - \rho}. \end{aligned}$$

Proof. We start from the normalized CE equation (B.28) and differentiate it with respect to s . This produces a first expression for dc_3/ds involving both

$w'_1(s)$ and the direct s -dependence of the outcomes.

Implicitly differentiating (B.28) with respect to s gives

$$c_3^s \left(\ln c_3 + \frac{s}{c_3} \frac{dc_3}{ds} \right) = w'_1(s) (\bar{y}^s - \underline{y}^s) + w_1(s) (\bar{y}^s \ln \bar{y} - \underline{y}^s \ln \underline{y}) + \underline{y}^s \ln \underline{y} \quad (\text{B.30})$$

From (B.24), the quotient rule gives:

$$w'_1(s) = \frac{(c_1^s \ln c_1 - \underline{x}_1^s \ln \underline{x}_1)(\bar{x}_1^s - \underline{x}_1^s) - (c_1^s - \underline{x}_1^s)(\bar{x}_1^s \ln \bar{x}_1 - \underline{x}_1^s \ln \underline{x}_1)}{(\bar{x}_1^s - \underline{x}_1^s)^2}$$

$$-w'_1(s) = \frac{w_1 \bar{x}_1^s \ln \bar{x}_1 + (1 - w_1) \underline{x}_1^s \ln \underline{x}_1 - c_1^s \ln c_1}{\bar{x}_1^s - \underline{x}_1^s}. \quad (\text{B.31})$$

Solving (B.30) for dc_3/ds :

$$\frac{dc_3}{ds} = \frac{1}{s c_3^{s-1}} \left[w'_1(s) (\bar{y}^s - \underline{y}^s) + w_1(s) (\bar{y}^s \ln \bar{y} - \underline{y}^s \ln \underline{y}) + \underline{y}^s \ln \underline{y} - c_3^s \ln c_3 \right] \quad (\text{B.32})$$

At this point the formula is exact but not yet useful for determining the sign of dc_3/ds . The key observation is that the bracketed terms can be recognized as Jensen gaps of the convex function $\phi(t) = t \ln t$, which are always positive and factor into a scale term (depending on outcome levels) and a one-dimensional kernel (depending only on the outcome ratio). This factorization is what ultimately reduces Condition B to a monotonicity argument.

Define the Jensen gap of $\phi(t) = t \ln t$ for lottery L_k :

$$G_k = w_1 \phi(\bar{x}_k^s) + (1 - w_1) \phi(\underline{x}_k^s) - \phi(c_k^s), \quad (\text{B.33})$$

where $c_k^s = w_1 \bar{x}_k^s + (1 - w_1) \underline{x}_k^s$. Since ϕ is strictly convex, $G_k > 0$ for all s . Using

$\phi(x^s) = x^s \ln(x^s) = s x^s \ln x$, the three non- w'_1 terms in the bracket of (B.32) are

$$w_1 \bar{y}^s \ln \bar{y} + (1 - w_1) \underline{y}^s \ln \underline{y} - c_3^s \ln c_3 = \frac{G_3}{s}. \quad (\text{B.34})$$

Similarly, from (B.31), the numerator of $-w'_1(s)$ is G_1/s and the denominator is $\bar{x}_1^s - \underline{x}_1^s$, so

$$-w'_1(s) = \frac{G_1}{s (\bar{x}_1^s - \underline{x}_1^s)}. \quad (\text{B.35})$$

Substituting into (B.32):

$$\frac{dc_3}{ds} = \frac{1}{s c_3^{s-1}} \left[\frac{-G_1(\bar{y}^s - \underline{y}^s)}{s (\bar{x}_1^s - \underline{x}_1^s)} + \frac{G_3}{s} \right]. \quad (\text{B.36})$$

The last step is to factor each Jensen gap into a scale term and a ratio term. This is where the proof becomes easier to interpret: once this factorization is done, the two lotteries enter only through their low/high outcome ratios encoded in ρ_1 and ρ_3 . Because $w_1 = w_1(s) \in (0, 1)$ and

$$c_1^s = \bar{x}_1^s [w_1 + (1 - w_1)\rho_1],$$

$$c_3^s = \bar{y}^s [w_1 + (1 - w_1)\rho_3].$$

the Jensen gaps admit the factorizations

$$G_1 = (\bar{x}_1^s - \underline{x}_1^s) H_{w_1}(\rho_1), \quad (\text{B.37})$$

$$G_3 = (\bar{y}^s - \underline{y}^s) H_{w_1}(\rho_3). \quad (\text{B.38})$$

Substituting (B.37) and (B.38) into (B.36) yields (B.29). Q.E.D.

It remains to show that the kernel difference $H_{w_1(s)}(\rho_3) - H_{w_1(s)}(\rho_1)$ in (B.29) is nonzero whenever $b \neq 0$. This follows if H_w is strictly monotone in ρ , since $b \neq 0$ changes the outcome ratio and therefore $\rho_3 \neq \rho_1$. In other words, the

derivative-sign question has now been reduced to a comparison of two outcome ratios passed through a one-dimensional kernel.

Lemma B.12 (Monotonicity of the Condition B kernel). *For each fixed $w \in (0, 1)$, the function $H_w(\rho)$ is strictly decreasing on $(0, \infty) \setminus \{1\}$.*

Proof. To show that H_w is strictly decreasing, it is enough to show that $H'_w(\rho) < 0$ for every $\rho \neq 1$. Because H_w is defined as a ratio, we will apply the quotient rule. We therefore begin by differentiating the numerator.

Write

$$H_w(\rho) = \frac{f_w(\rho)}{1 - \rho},$$

where

$$f_w(\rho) = (1 - w)\rho \ln \rho - [w + (1 - w)\rho] \ln[w + (1 - w)\rho].$$

Differentiating f_w term by term gives

$$\begin{aligned} \frac{d}{d\rho} [(1-w)\rho \ln \rho] &= (1-w)(\ln \rho + 1), \\ \frac{d}{d\rho} \{ -[w + (1-w)\rho] \ln[w + (1-w)\rho] \} &= -(1-w)(\ln[w + (1-w)\rho] + 1). \end{aligned}$$

The constant terms $\pm(1-w)$ cancel, leaving

$$f'_w(\rho) = (1-w) \ln \frac{\rho}{w + (1-w)\rho}. \quad (\text{B.39})$$

The sign of $f'_w(\rho)$ changes at $\rho = 1$, so f'_w alone is not enough to establish monotonicity of H_w on the full domain. We therefore return to H_w itself and apply the quotient rule.

$$H'_w(\rho) = \frac{f'_w(\rho)(1 - \rho) + f_w(\rho)}{(1 - \rho)^2}. \quad (\text{B.40})$$

Because the denominator is positive for $\rho \neq 1$, the sign of H'_w is the sign of the numerator

$$f'_w(\rho)(1 - \rho) + f_w(\rho).$$

So the problem is now reduced to showing that this numerator is negative whenever $\rho \neq 1$. At $\rho = 1$, it equals zero because $f_w(1) = 0$ and $1 - \rho = 0$.

The next step is to understand how this numerator behaves as ρ moves away from 1. Differentiating it once more gives

$$\frac{d}{d\rho} [f'_w(\rho)(1 - \rho) + f_w(\rho)] = f''_w(\rho)(1 - \rho). \quad (\text{B.41})$$

So everything now depends on the sign of $f''_w(\rho)$. From (B.39) we have

$$f''_w(\rho) = (1-w) \left[\frac{1}{\rho} - \frac{1-w}{w+(1-w)\rho} \right] = \frac{w(1-w)}{\rho[w+(1-w)\rho]} > 0. \quad (\text{B.42})$$

Therefore the numerator of $H'_w(\rho)$ has derivative $f''_w(\rho)(1 - \rho)$, which is positive when $\rho < 1$ and negative when $\rho > 1$. In other words, that numerator increases up to $\rho = 1$ and then decreases afterward. Since it equals zero at $\rho = 1$, it follows that $\rho = 1$ is its unique maximum, so the numerator is strictly negative for every $\rho \neq 1$. Going back to (B.40), this implies $H'_w(\rho) < 0$ for all $\rho \neq 1$, and therefore H_w is strictly decreasing on $(0, \infty) \setminus \{1\}$. Q.E.D.

This is exactly the property needed for Condition B: once $b \neq 0$ changes the low/high outcome ratio, the kernel terms cannot cancel, so the derivative of the residual CE equation cannot vanish away from the log-utility limit.

Proposition B.13 (Shared-probability Condition B for CRRA). *Under the shared-probability design $p_3 = p_1$, Condition B holds for all $s \neq 0$ if and only if $b \neq 0$, equivalently $\bar{x}_1/\underline{x}_1 \neq \bar{x}_3/\underline{x}_3$.*

Proof. The case $b = 0$ was handled in Step 1: then $c_3 = m c_1$ is independent of s , so Condition B fails. It remains to prove sufficiency.

Assume $b \neq 0$, equivalently $e \neq 0$. Since

$$\underline{y} = \underline{x}_1 + e,$$

$$\bar{y} = \bar{x}_1 + e.$$

we have

$$\frac{\underline{y}}{\bar{y}} - \frac{\underline{x}_1}{\bar{x}_1} = \frac{e(\bar{x}_1 - \underline{x}_1)}{\bar{x}_1(\bar{x}_1 + e)} \neq 0.$$

So the normalized lottery has a different low/high outcome ratio from L_1 .

Now fix any $s \neq 0$. By Lemma B.11, dc_3/ds is the product of two terms:

$$\frac{\bar{y}^s - \underline{y}^s}{s^2 c_3^{s-1}} \quad \text{and} \quad H_{w_1(s)}(\rho_3) - H_{w_1(s)}(\rho_1).$$

So it is enough to show that both terms are nonzero.

Start with the kernel difference. The map $z \mapsto z^s$ is one-to-one on $(0, \infty)$, so

$$\begin{aligned} \rho_3 &= \left(\frac{\underline{y}}{\bar{y}} \right)^s \\ &\neq \left(\frac{\underline{x}_1}{\bar{x}_1} \right)^s \\ &= \rho_1 \end{aligned}$$

for every $s \neq 0$.

By Lemma B.12, $H_{w_1(s)}$ is strictly decreasing in ρ , so $\rho_3 \neq \rho_1$ implies

$$H_{w_1(s)}(\rho_3) - H_{w_1(s)}(\rho_1) \neq 0.$$

Now consider the prefactor. Since $0 < \underline{y} < \bar{y}$, the difference $\bar{y}^s - \underline{y}^s$ has the same sign as s , and $s^2 c_3^{s-1} > 0$, so the prefactor is nonzero for every $s \neq 0$.

Both terms in (B.29) are therefore nonzero, so $dc_3/ds \neq 0$. In fact, its sign is constant across the entire domain. To see this, let $q_1 = \underline{x}_1/\bar{x}_1$ and $q_3 = \underline{y}/\bar{y}$. Since $q_3 - q_1$ has the sign of e , the map $z \mapsto z^s$ implies that $\rho_3 - \rho_1$ has the sign of e when $s > 0$ and the opposite sign when $s < 0$. Because $H_{w_1(s)}$ is strictly decreasing, the kernel difference $H_{w_1(s)}(\rho_3) - H_{w_1(s)}(\rho_1)$ reverses that sign. The prefactor has the sign of s , as noted above. Combining these,

$$\text{sign}\left(\frac{dc_3}{ds}\right) = -\text{sign}(e)$$

for every $s \neq 0$.

Thus $c_3(s)$ is strictly monotone on each branch $(-\infty, 0)$ and $(0, \infty)$, with the same direction on both.

To connect the two branches, note that

$$w_1(s) = \frac{c_1^s - \underline{x}_1^s}{\bar{x}_1^s - \underline{x}_1^s}$$

has the finite limit

$$w_1(0) = \lim_{s \rightarrow 0} w_1(s) = \frac{\ln c_1 - \ln \underline{x}_1}{\ln \bar{x}_1 - \ln \underline{x}_1} \in (0, 1),$$

by L'Hôpital's rule. Writing

$$c_3(s)^s = w_1(s) \bar{y}^s + (1 - w_1(s)) \underline{y}^s,$$

the expansion $z^s = 1 + s \ln z + o(s)$ gives

$$c_3(s)^s = 1 + s \left[w_1(0) \ln \bar{y} + (1 - w_1(0)) \ln \underline{y} \right] + o(s),$$

so

$$\lim_{s \rightarrow 0} c_3(s) = \bar{y}^{w_1(0)} \underline{y}^{1-w_1(0)}.$$

Thus c_3 extends continuously through $s = 0$. Because the derivative has the same sign on both branches, this common limit separates their images. When $e > 0$, the derivative $dc_3/ds < 0$ for all $s \neq 0$, so $c_3(s) > c_3(0)$ on $(-\infty, 0)$ and $c_3(s) < c_3(0)$ on $(0, \infty)$. The case $e < 0$ is symmetric, with the inequalities reversed. In either case, the two branches have disjoint images, so no value can be attained on both. Therefore Condition B holds for all $s \neq 0$, equivalently for all $r \neq 1$.

Finally, $b = 0$ if and only if $\bar{x}_1/\underline{x}_1 = \bar{x}_3/\underline{x}_3$. One direction is immediate from (B.21). For the converse, if the ratios are equal then

$$\begin{aligned} \frac{m\bar{x}_1 + b}{m\underline{x}_1 + b} &= \frac{\bar{x}_1}{\underline{x}_1} \\ &\iff b\underline{x}_1 = b\bar{x}_1 \\ &\iff b(\bar{x}_1 - \underline{x}_1) = 0, \end{aligned}$$

and $\bar{x}_1 \neq \underline{x}_1$.

Q.E.D.

B.3.3 Full three-parameter identification

Combining the results: the W -bijectivity of Section B.2.2 and the diagonal L -matrix establish Condition A (Proposition B.10) for any $p_1 \neq p_2$. The shared-probability design $p_3 = p_1$ with $b \neq 0$ (equivalently $\bar{x}_1/\underline{x}_1 \neq \bar{x}_3/\underline{x}_3$) establishes Condition B (Proposition B.13) on the power-CRRA domain $\mathbb{R}_{>0} \setminus \{1\}$, equivalently $s = 1 - r \neq 0$. Thus the CE map $F: (r, w_1, w_2) \mapsto (c_1, c_2, c_3)$ is injective on the full-measure subset

$$((0, \infty) \setminus \{1\}) \times W(\mathbb{R}_{>0}^2).$$

The excluded singleton $r = 1$ corresponds to the log-utility limit and is immaterial for the Bayesian change-of-variables prior.

B.3.4 Condition C: noise identification

The main text uses a logistic Fechner model with $\Lambda(t) = 1/(1+e^{-t})$. Condition C (Section B.1.4) holds for any choice task with $\Delta V \neq 0$. For a non-degenerate design, the indifference locus $\Delta V = 0$ is a lower-dimensional exceptional set, so combined with Conditions A and B the full map $\tilde{F}: (r, w_1, w_2, \lambda) \mapsto (c_1, c_2, c_3, P_4)$ is injective on the full-measure subset relevant for Bayesian estimation. This is exactly the level of injectivity needed to justify the transformed prior in the main text.

B.4 Extensions and Discussion

The sequential approach scales linearly: q Block I lotteries plus p Block II lotteries, plus one choice task for λ . The all-binary design handles Block I for $q = 2$ (covering Prelec, Goldstein–Einhorn, Tversky–Kahneman); Block II lotteries are chosen freely to provide outcome-side variation. Condition C requires one additional choice task with $\Delta V \neq 0$. For utility families with $p > 1$ (e.g., expo-power), Condition B becomes a $p \times p$ injectivity condition. In those cases the scalar derivative-sign argument used for CRRA is replaced by a higher-dimensional reduced system, but the same sequential decomposition into Block I, Block II, and the noise task still applies.

Appendix C: Identification from Choice Probabilities under EUT

The quantity-space prior specification of Section 3 requires an injective mapping from parameters to predictive quantities, and the Jacobian pullback must faithfully recover the intended prior on the feasible image. As noted in Section 6, certainty equivalents require inverting the utility function, which complicates the injectivity proof for richer utility families. Choice probabilities avoid this complication.

This appendix proves that, under Expected Utility Theory with a logit choice rule, the Jacobian pullback from choice probabilities recovers the intended prior for any analytic utility family, provided the lottery design satisfies a rank condition that is easy to verify.

The key observation is that the map from p utility parameters to p pairwise choice probabilities factors as $G = \Lambda \circ (C\phi + \mathbf{d})$, where ϕ evaluates utility at interior outcomes, C is a contrast matrix determined by the lottery design, and Λ is the coordinatewise logistic link. Since Λ is a bijection with strictly positive derivative, all identification content lies in the product $C D\phi$. The results here apply to EUT specifically; extending to non-expected utility functionals such as RDU, where the probability weighting layer breaks the affine structure of $C\phi + \mathbf{d}$, is left for future work. In that sense, this appendix complements Appendix B: Appendix B handles a richer RDU specification using certainty equivalents, while the present appendix trades that richer behavioral structure for a cleaner design-side factorization in choice-probability space.

C.1 Setup

Let $\Theta \subseteq \mathbb{R}^p$ be open and bounded, and let $\{u(\cdot; \boldsymbol{\theta})\}_{\boldsymbol{\theta} \in \Theta}$ be a family of utility functions on $[x_{\min}, x_{\max}]$ such that for each $\boldsymbol{\theta}$, $x \mapsto u(x; \boldsymbol{\theta})$ is strictly increasing with $u(x_{\min}; \boldsymbol{\theta}) = 0$ and $u(x_{\max}; \boldsymbol{\theta}) = 1$. Assume u is analytic in $\boldsymbol{\theta}$ and C^1 in $(x, \boldsymbol{\theta})$.

We observe $R = p$ pairwise comparisons. Across all comparisons, let

$$x_{\min} = y_0 < m_1 < \cdots < m_M < y_{M+1} = x_{\max} \quad (\text{C.1})$$

be the common support obtained by taking the distinct outcomes that appear

anywhere in the design and adjoining the normalization endpoints if necessary. For comparison r , let $w_{r\ell} := \Pr_{A_r}(y_\ell) - \Pr_{B_r}(y_\ell)$ be the signed probability contrast at outcome y_ℓ . Under EUT, the expected-utility difference is

$$\Delta_r(\boldsymbol{\theta}) = \sum_{\ell=0}^{M+1} w_{r\ell} u(y_\ell; \boldsymbol{\theta}). \quad (\text{C.2})$$

Define the *evaluation map*

$$\phi(\boldsymbol{\theta}) = \left(u(m_1; \boldsymbol{\theta}), \dots, u(m_M; \boldsymbol{\theta}) \right) \in (0, 1)^M, \quad (\text{C.3})$$

which records utility at the M interior outcomes. Since $u(x_{\min}; \boldsymbol{\theta}) = 0$ kills the $\ell = 0$ term and $u(x_{\max}; \boldsymbol{\theta}) = 1$ reduces the $\ell = M+1$ term to the contrast weight alone, (C.2) simplifies to

$$\Delta(\boldsymbol{\theta}) = C \phi(\boldsymbol{\theta}) + \mathbf{d}, \quad (\text{C.4})$$

where $C = (w_{ri})$ is the $p \times M$ *contrast matrix* and $d_r = w_{r, M+1}$. The observed choice probability is $\pi_r(\boldsymbol{\theta}) = \Lambda(\Delta_r(\boldsymbol{\theta}))$ with $\Lambda(t) = e^t / (1 + e^t)$, giving the *choice-probability map*

$$G(\boldsymbol{\theta}) = \left(\pi_1(\boldsymbol{\theta}), \dots, \pi_p(\boldsymbol{\theta}) \right) = \Lambda(C \phi(\boldsymbol{\theta}) + \mathbf{d}). \quad (\text{C.5})$$

Normalization matters here in a way it does not for certainty equivalents. For certainty equivalents, positive affine transformations of utility are innocuous. For logit probabilities they are not: additive shifts cancel in Δ_r , but rescalings do not. Without endpoint normalization, the utility scale would be confounded with the logit noise scale.

C.2 Identification

Theorem C.1 (Identification from p choice probabilities). *Let $G: \Theta \rightarrow (0, 1)^p$ be the choice-probability map (C.5).*

1. *Jacobian. $J_G(\boldsymbol{\theta}) = D_\Lambda(\boldsymbol{\theta}) C D\phi(\boldsymbol{\theta})$, where $D_\Lambda = \text{diag}(\pi_r(1 - \pi_r))_{r=1}^p$, which is a positive diagonal matrix since each $\pi_r \in (0, 1)$. Hence*

$$\text{rank}(J_G(\boldsymbol{\theta})) = \text{rank}(C D\phi(\boldsymbol{\theta})). \quad (\text{C.6})$$

2. *Singular set. At some parameter values the Jacobian may lose rank, so that G is not locally injective there. Call these the singular set $S = \{\boldsymbol{\theta} : \text{rank}(J_G) < p\}$. If $\text{rank}(C D\phi(\boldsymbol{\theta}_0)) = p$ at even one point $\boldsymbol{\theta}_0 \in \Theta$, then S has Lebesgue measure zero and $\Theta \setminus S$ is open and dense. In other words, rank failure is the exception, not the rule. It can occur, but only on a negligible set of parameter values.*

3. *Local identification. At every $\boldsymbol{\theta} \in \Theta \setminus S$ the Jacobian has full rank, so G is locally injective and nearby parameter values produce distinct choice probabilities.*

4. *Bayesian pullback. Let q be a prior density on $(0, 1)^p$. If, for almost every $\boldsymbol{\pi} \in G(\Theta)$, the equation $G(\boldsymbol{\theta}) = \boldsymbol{\pi}$ has exactly m solutions, then the pullback*

$$\tilde{q}(\boldsymbol{\theta}) \propto q(G(\boldsymbol{\theta})) |\det J_G(\boldsymbol{\theta})| \quad (\text{C.7})$$

induces the density $m q(\boldsymbol{\pi})$ on $G(\Theta)$. After normalization, this is exactly the intended prior restricted to the feasible image $G(\Theta)$. Any prior mass that q assigns outside $G(\Theta)$ is irrelevant because no parameter value can

reach it. When $m = 1$, no multiplicity correction is needed.

Proof. Part 1. Since $G = \Lambda \circ (C\phi + \mathbf{d})$ and $\Lambda'(t) = \Lambda(t)(1 - \Lambda(t)) > 0$, the chain rule gives $J_G = D_\Lambda C D\phi$. Since D_Λ is a positive diagonal matrix, multiplying by it does not change rank.

Part 2. Whether J_G has full rank at $\boldsymbol{\theta}$ depends, by Part 1, on whether $C D\phi(\boldsymbol{\theta})$ has full rank, which in the square case ($M = p$) amounts to whether $\det(C D\phi(\boldsymbol{\theta}))$ is nonzero. Write $f(\boldsymbol{\theta}) = \det(C D\phi(\boldsymbol{\theta}))$. Because u is analytic in $\boldsymbol{\theta}$, the function f is analytic. Because there exists $\boldsymbol{\theta}_0$ such that $f(\boldsymbol{\theta}_0) \neq 0$, the function f is not identically zero. A standard property of analytic functions is that a nonzero analytic function can only vanish on a set of Lebesgue measure zero (Krantz and Parks 2002; Mityagin 2015). Therefore $S = \{f = 0\}$ has measure zero. When $M > p$, the same argument applies to any $p \times p$ subdeterminant of $C D\phi$ that is nonzero at $\boldsymbol{\theta}_0$, and S is contained in its zero set.

Part 3. At each $\boldsymbol{\theta} \in \Theta \setminus S$, $J_G(\boldsymbol{\theta})$ has full rank p . By the inverse function theorem, G is locally injective at such $\boldsymbol{\theta}$, and no nearby parameter value produces the same choice probabilities.

Part 4. Since Λ is strictly monotone, $G(\boldsymbol{\theta}_1) = G(\boldsymbol{\theta}_2)$ if and only if $C\phi(\boldsymbol{\theta}_1) + \mathbf{d} = C\phi(\boldsymbol{\theta}_2) + \mathbf{d}$. So G has the same number of preimages at each point as the utility-difference map $C\phi + \mathbf{d}$. The multivariate change-of-variables formula then gives, for every measurable $B \subseteq G(\Theta)$,

$$\int_{G^{-1}(B)} q(G(\boldsymbol{\theta})) |\det J_G(\boldsymbol{\theta})| d\boldsymbol{\theta} = \int_B q(\boldsymbol{\pi}) \#G^{-1}(\boldsymbol{\pi}) d\boldsymbol{\pi} = m \int_B q(\boldsymbol{\pi}) d\boldsymbol{\pi}. \quad (\text{C.8})$$

Q.E.D.

Part 4 is the key Bayesian consequence. It shows that the researcher's quantity-space prior is recovered on the feasible image even when the forward map is only

finitely many-to-one, provided the multiplicity is constant.

Identification requires $M \geq p$ interior outcomes and $\text{rank}(C) \geq p$. If any lottery pair has A_r first-order stochastically dominating B_r , the corresponding Δ_r is nonnegative for every increasing utility, so that comparison fixes the sign of the EU gap. Such comparisons can still be informative about utility parameters through the magnitude of Δ_r , so non-FOSD is not necessary for identification. It is, however, often a useful design feature because non-FOSD pairs allow the sign of the EU gap itself to vary with preferences. In any case, one still needs $C D\phi$ to have rank p .

C.3 Sufficient Designs

The theorem applies to any lottery design whose contrast matrix C and evaluation Jacobian $D\phi$ jointly produce rank p . Two examples illustrate this.

Sure amount versus endpoint lottery. Fix interior outcomes $m_1, \dots, m_p \in (x_{\min}, x_{\max})$ and $\lambda_1, \dots, \lambda_p \in (0, 1)$. For each i , compare a sure amount m_i against a lottery paying $x_{\max}$ with probability λ_i and $x_{\min}$ otherwise. Normalization gives $\text{EU}(B_i) = \lambda_i$, so $\Delta_i(\theta) = u(m_i; \theta) - \lambda_i$. The only interior outcome in pair i is m_i , with contrast weight $+1$, so $C = I_p$ and the Jacobian simplifies to

$$J_G(\theta) = \text{diag}(\pi_i(1 - \pi_i))_{i=1}^p \cdot \left(\frac{\partial u(m_i; \theta)}{\partial \theta_j} \right)_{i,j}. \quad (\text{C.9})$$

With this design, identification reduces entirely to the evaluation map: the utility family must be such that the matrix of partial derivatives $(\partial u(m_i; \theta) / \partial \theta_j)_{i,j}$ has full rank at some θ_0 . For any concrete design, this rank condition can be checked numerically at a single convenient parameter value before estimation.

Risk-apportionment tasks. A risk-apportionment (RA) task of order k is a

binary lottery comparison in which the first $k - 1$ moments of the two lotteries are matched and the k th moments differ (Eeckhoudt and Schlesinger 2006). For example, RA2 (risk aversion) matches means but differs in variance, RA3 (prudence) matches means and variances but differs in skewness, and so on. Each RA task contributes a row to C . Unlike the binary design, C is no longer the identity, so the design-side rank condition is not automatic. However, it is straightforward to verify numerically for any given set of RA tasks. RA tasks are a natural choice because each task targets a different order of the utility function, so for utility families whose parameters control these derivatives somewhat independently, the rows of $C D\phi$ will generically point in different directions.

C.4 Noise Parameter and Sequential Composition

The results above identify p utility parameters from p choice probabilities. In practice, the choice model also includes a noise parameter λ that scales the logit link. This can be handled by sequential composition: first identify the p utility parameters using p lottery pairs as above, then identify λ with one additional pair whose choice probability varies with λ given the utility parameters. The $(p+1)$ th pair contributes one additional row to the system, and the same Jacobian pullback applies to the extended $(p + 1)$ -dimensional map.

Appendix D: Sensitivity to (α, κ)

This appendix reports results from re-estimating the CE-space Beta-Beta model at three additional (α, κ) configurations beyond the baseline ($\alpha = 0.2$, $\kappa = 0.25$). The four configurations span the corners of the interpretable region:

Configuration	α	κ
Baseline	0.2	0.25
High κ	0.2	1.50
High α	0.8	0.25
Both high	0.8	1.50

Low α (0.2) allocates most direct prior variance to within-population heterogeneity; high α (0.8) attributes most of it to uncertainty about the population mean. Low κ (0.25) expresses high confidence in the heterogeneity level; high κ (1.50) expresses substantial uncertainty.

All four configurations use the same direct prior (mean and variance for each CE and the choice probability), the same data, the same Beta-Beta distributional family, and the same diffuse NIG comparison prior used in the main text. The only thing that changes is the hierarchical allocation of prior variance.

Configuration	α	κ	Well	Moderate	Dominated	Dominated (%)
Baseline	0.2	0.25	80	13	18	16
High κ	0.2	1.50	83	14	14	13
High α	0.8	0.25	76	14	21	19
Both high	0.8	1.50	81	13	17	15

The table makes the main pattern clear. Raising α increases the share of prior-dominated subjects from 16% in the baseline to 19% when $\alpha = 0.8$ and $\kappa = 0.25$, consistent with allocating more uncertainty to population-mean variation. Raising κ alone lowers the prior-dominated share to 13%, reflecting the weaker concentration of the hyperprior over heterogeneity. The figures below

show the same pattern from three complementary angles: CDF differences, posterior predictive CE overlays, and the induced parameter-space prior.

D.1 CDF Difference Plots

The following figures show, for each configuration, the difference between the informative-posterior and diffuse-posterior CDFs of each subject’s CE, with one curve per subject and one panel per reference lottery. Read together, they separate the roles of κ and α before showing how the two interact when both are large.

With κ raised to 1.50 while α remains at 0.2, the hyperprior over heterogeneity is substantially less concentrated. Relative to the baseline, the CDF differences are slightly attenuated rather than qualitatively changed, which matches the lower prior-dominated share reported in the table above. In other words, increasing κ mainly weakens the strength of the regularization without changing where it shows up.

By contrast, raising α to 0.8 allocates the majority of direct prior variance to uncertainty about the population mean, leaving less room for individual heterogeneity. This produces the highest prior-dominated share across the four configurations, with larger CDF deviations for the same prior-sensitive subjects. So the main effect of higher α is not to create new problem cases, but to amplify prior influence for subjects who were already weakly identified.

When both α and κ are high, the effects partially offset. The large α concentrates heterogeneity while the large κ expresses uncertainty about that concentrated specification, returning the prior-dominated share close to the baseline level. This final panel is therefore useful mainly as an interaction check: it shows that

Figure D.1: CDF differences: $\alpha = 0.2$, $\kappa = 1.50$

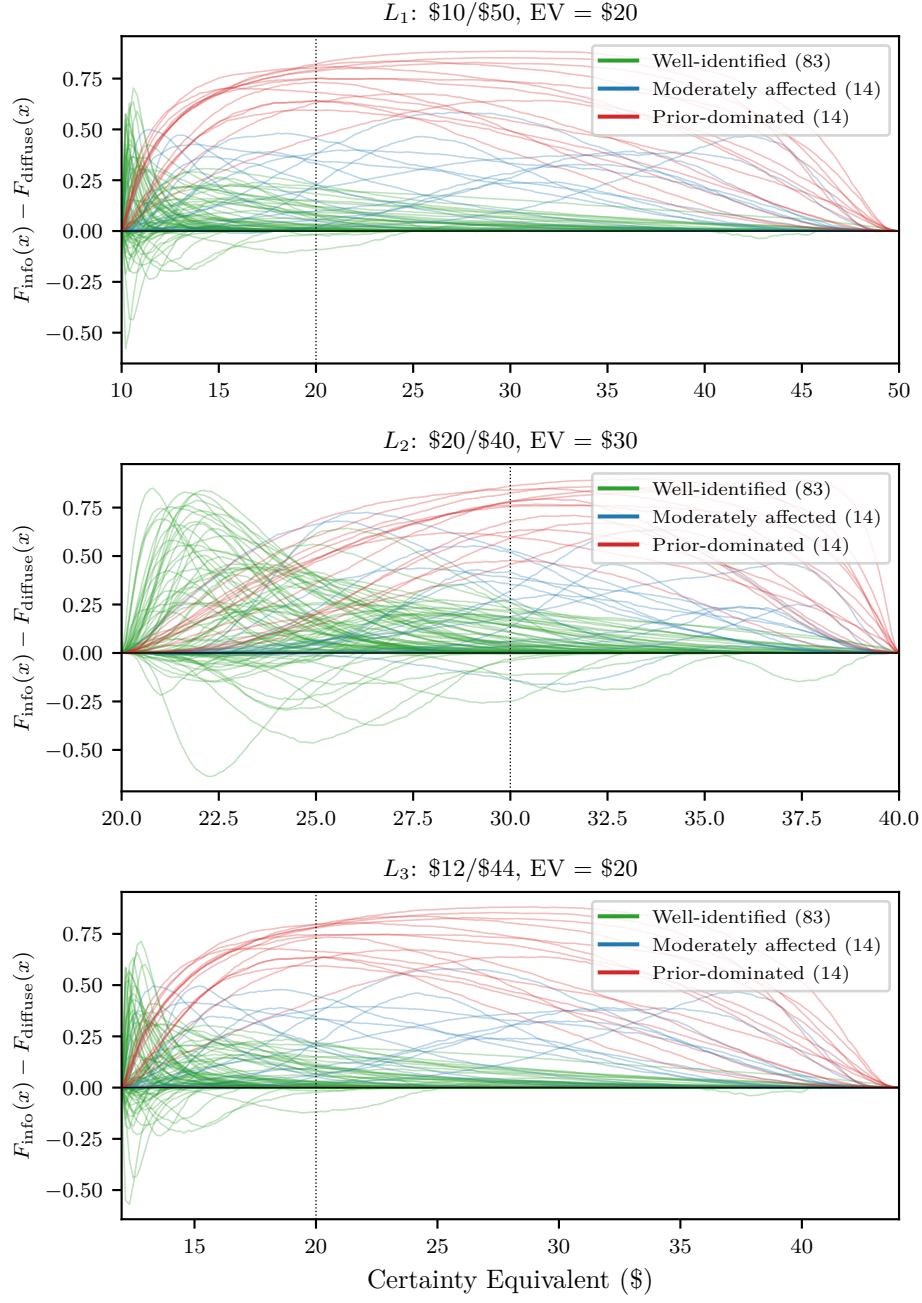


Figure D.2: CDF differences: $\alpha = 0.8$, $\kappa = 0.25$

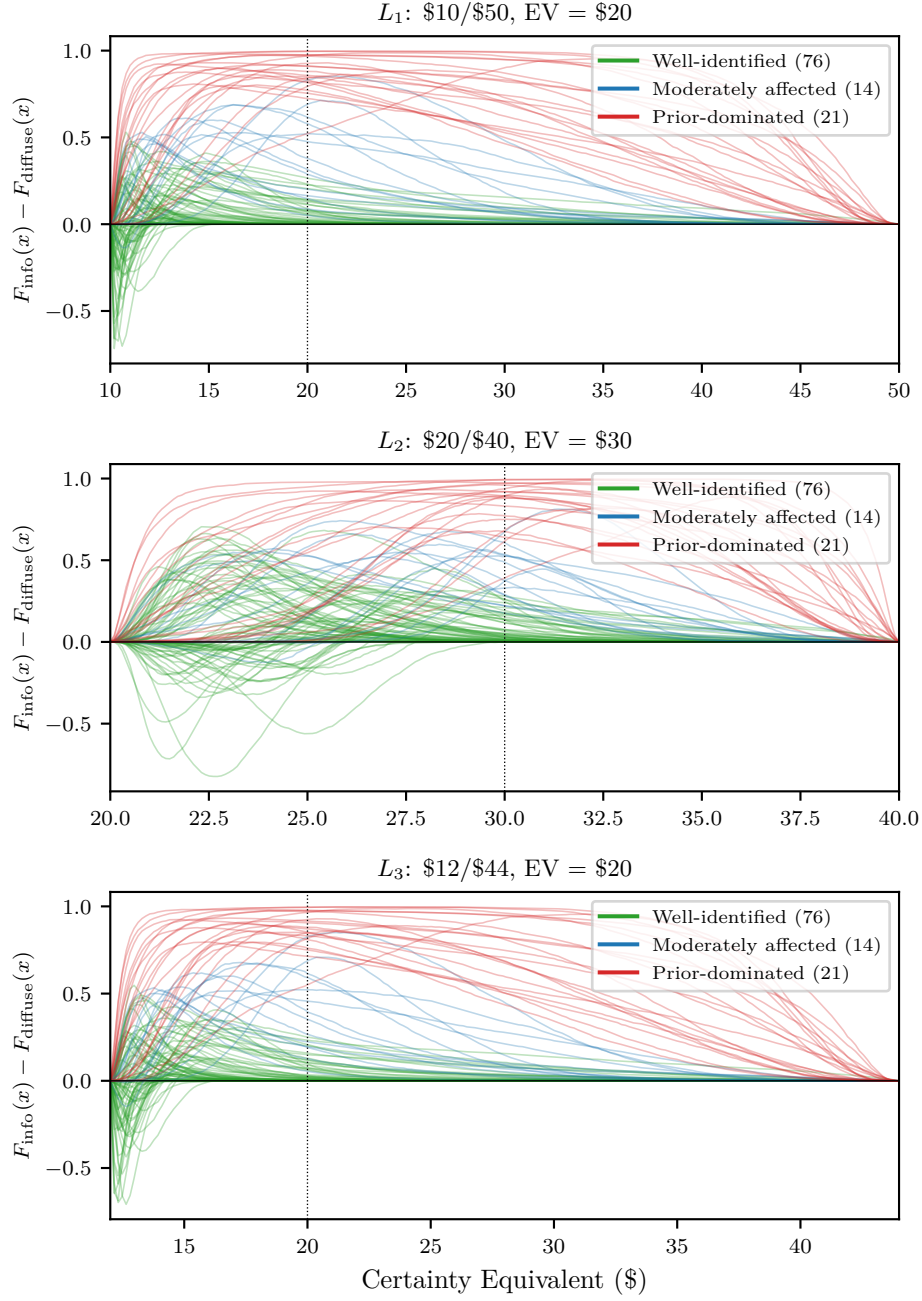
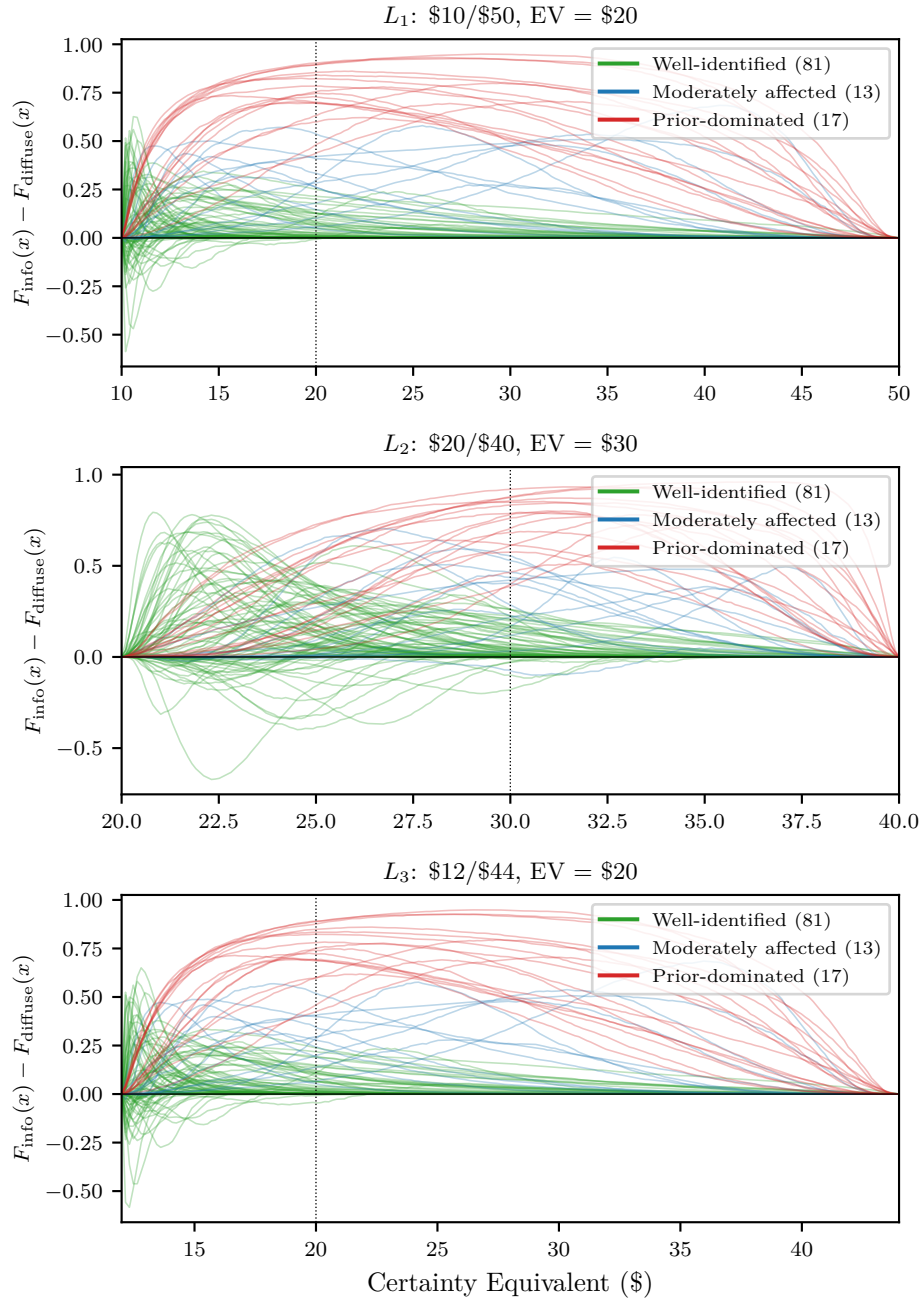


Figure D.3: CDF differences: $\alpha = 0.8$, $\kappa = 1.50$



the separate directional effects of α and κ remain visible, but neither dominates completely once both are moved away from the baseline.

D.2 Predictive CE Overlays

The following figures compare the hyper-posterior predictive CE distributions across configurations. They present the same sensitivity exercise in levels rather than CDF differences, making it easier to see whether the substantive ordering of lottery evaluations changes as (α, κ) moves.

Across all three alternatives, the overlay figures show that changing (α, κ) mostly affects dispersion and shrinkage rather than qualitative ordering. The same lottery-specific pattern remains visible across configurations, with the largest differences concentrated in the wider-range lotteries where the informative prior has more room to move posterior mass. This is the level-space counterpart to the CDF plots above: prior influence changes in magnitude, but the substantive ranking of posterior predictive CEs remains stable.

D.3 Induced Prior in Parameter Space

The quantity-space framework specifies priors on predictive quantities and lets the Jacobian correction induce the parameter-space prior. To visualize this induced prior, I sample from the full hierarchical model with no data likelihood, so that the resulting parameter draws reflect only the CE-space specification and the Jacobian geometry.

Figure D.7 shows the result. The induced prior on r is centered near mild risk aversion (median 1.08) with support for both risk-seeking and strongly risk-averse individuals. The Prelec curvature ϕ peaks near 1 (linear probability

Figure D.4: Predictive CE overlay: $\alpha = 0.2$, $\kappa = 1.50$

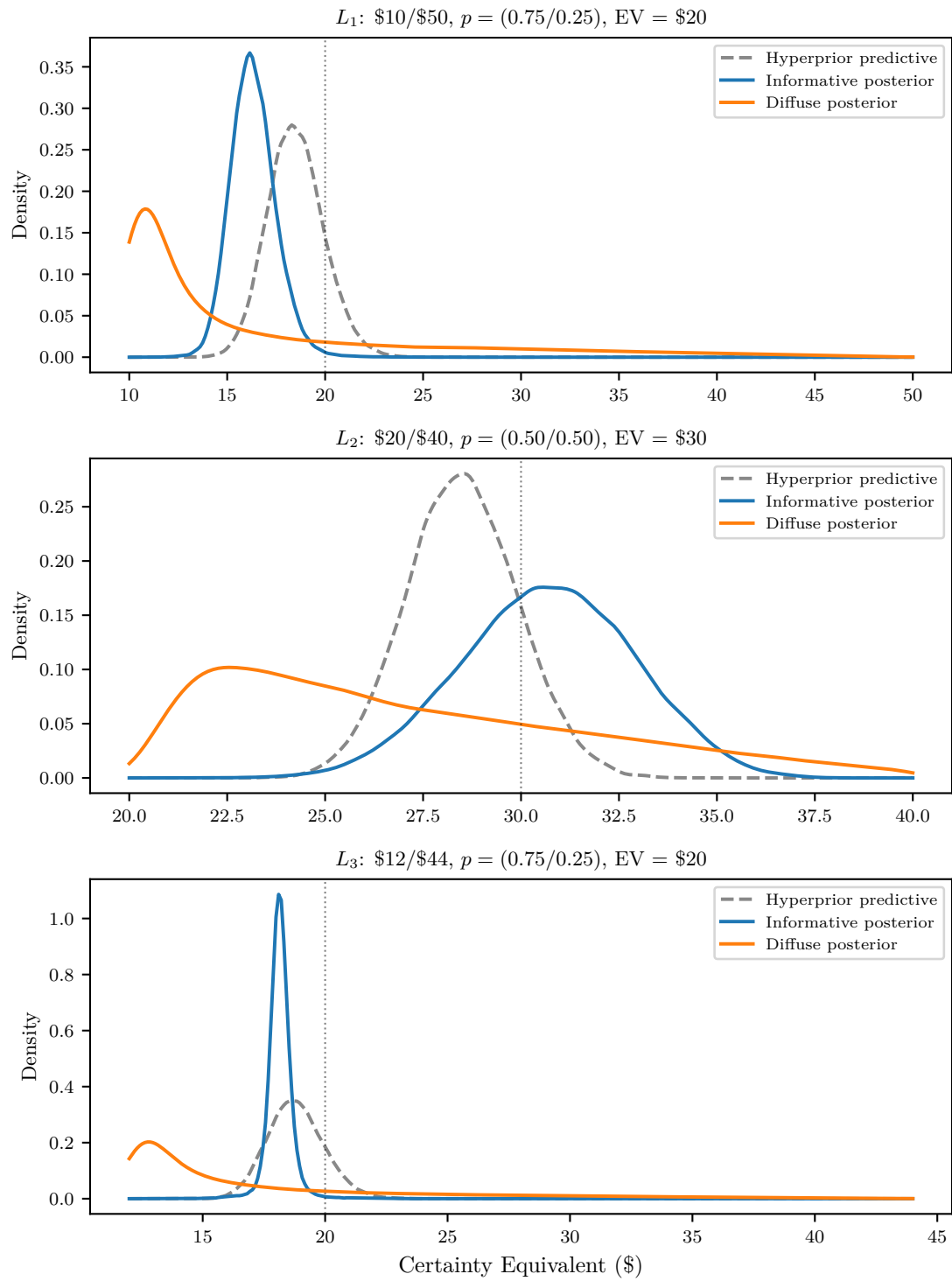


Figure D.5: Predictive CE overlay: $\alpha = 0.8$, $\kappa = 0.25$

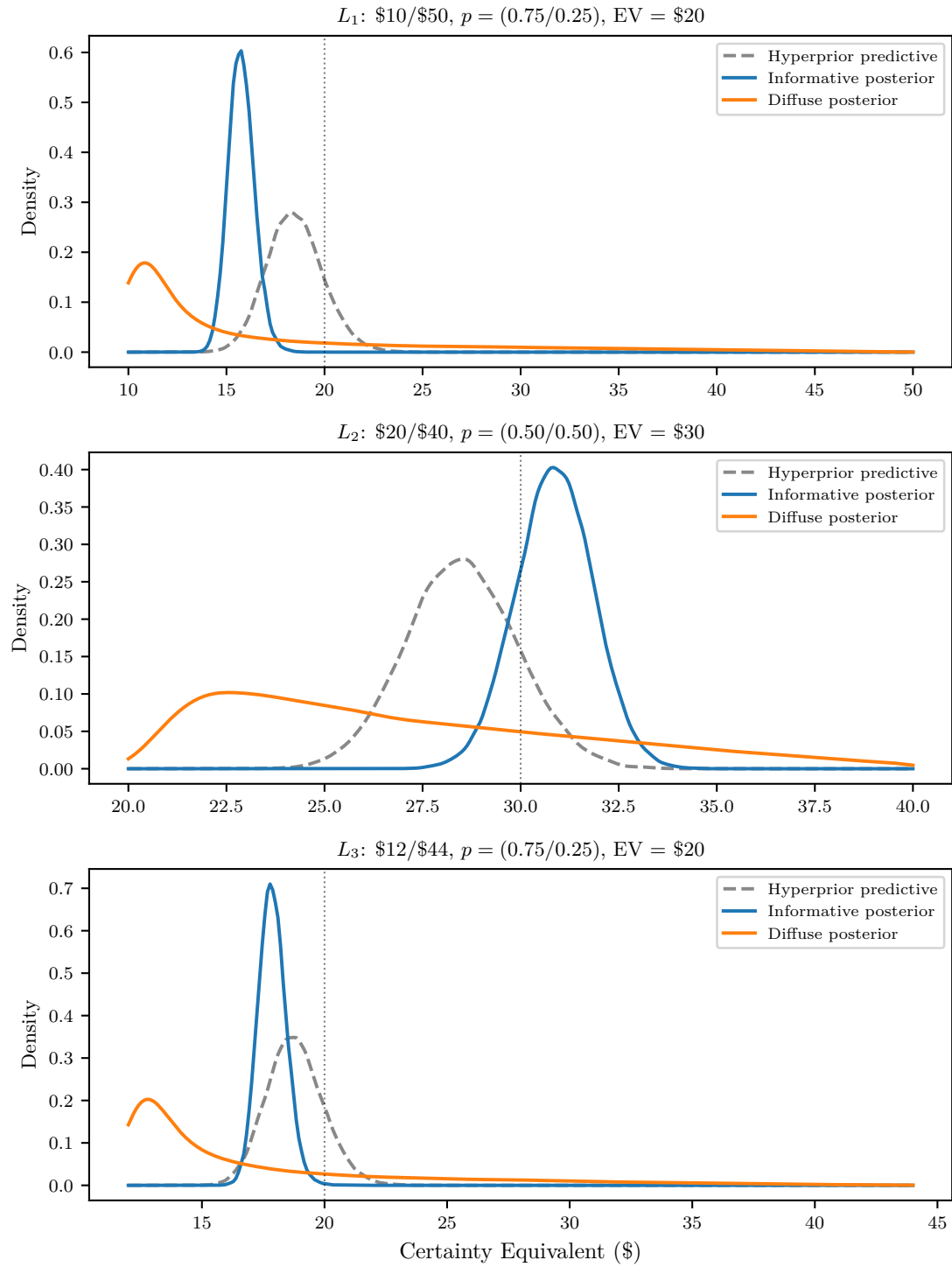


Figure D.6: Predictive CE overlay: $\alpha = 0.8$, $\kappa = 1.50$

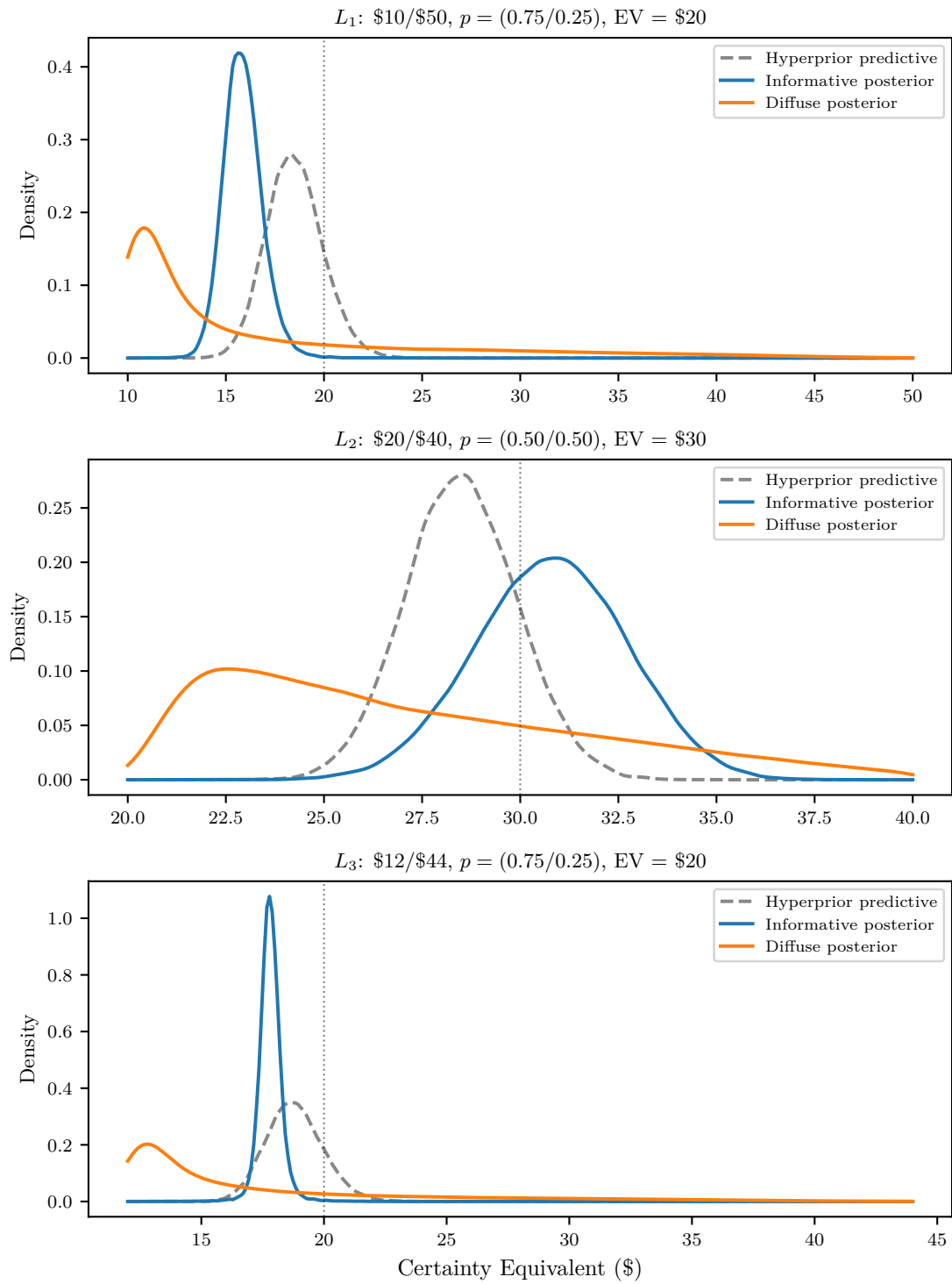
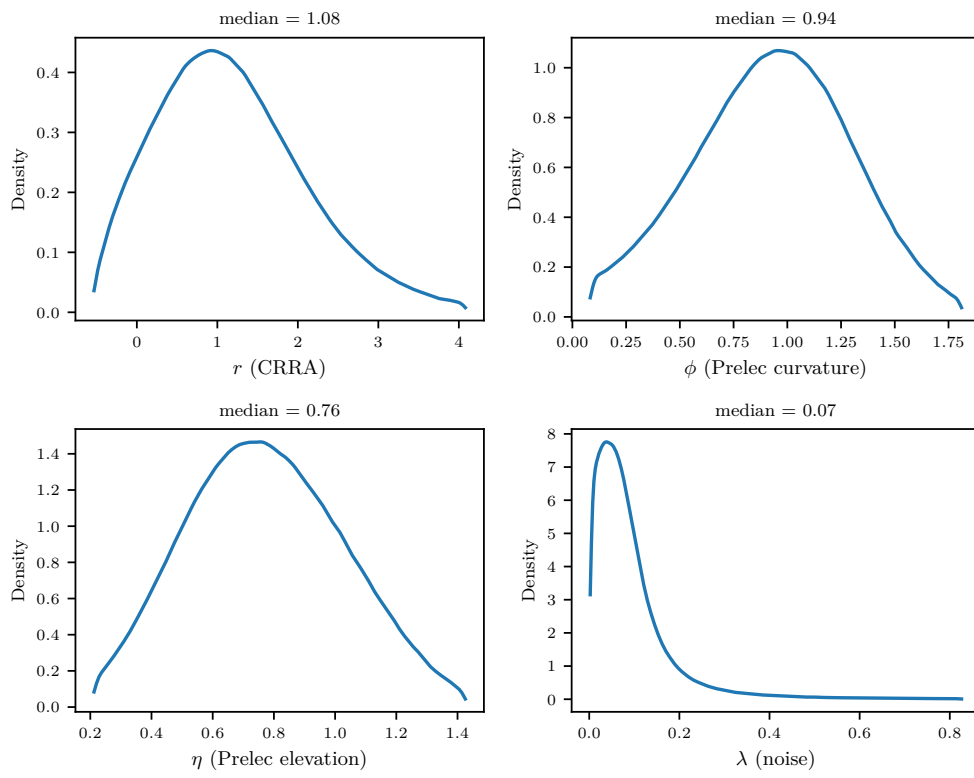


Figure D.7: Hyperprior predictive distribution in parameter space



weighting), allowing inverse-S shapes ($\phi < 1$) that are typical of the empirical literature. The elevation η concentrates below 1, and the noise λ is concentrated near zero with a long right tail. These are sensible parameter-space priors for an RDU model, achieved entirely through beliefs about certainty equivalents and a choice probability.

D.4 Structural Parameter Posteriors for Selected Subjects

The following figures show how data updates the induced prior for individual subjects. Together they connect the sensitivity patterns above back to parameter space: Figure D.8 shows a subject for whom the prior meaningfully shapes the posterior, while Figure D.9 shows a subject for whom the likelihood largely overwhelms it.

For the most affected subject, the two prior specifications recover fundamentally different parameter regions. The informative model concentrates on high r (strong risk aversion), low ϕ (strong inverse-S probability weighting), and low η , while the diffuse model favors negative r (risk seeking) with different probability weighting parameters. These are the same opposing modes visible in the CE comparison, now shown in parameter space. The noise parameter λ is similar across models, confirming that the disagreement is about preferences rather than noise.

For the least affected subject, the posteriors are nearly identical across models. The data are sufficiently informative that the prior specification has negligible influence on the structural parameter estimates.

Taken together, these sensitivity checks reinforce the main text’s interpretation of (α, κ) . Changing the variance allocation and the researcher’s confidence in that allocation does move the induced prior and, for some weakly identified subjects, can shift posterior inferences in noticeable ways. But the broad qualitative

Figure D.8: Posterior parameter distributions: most affected subject

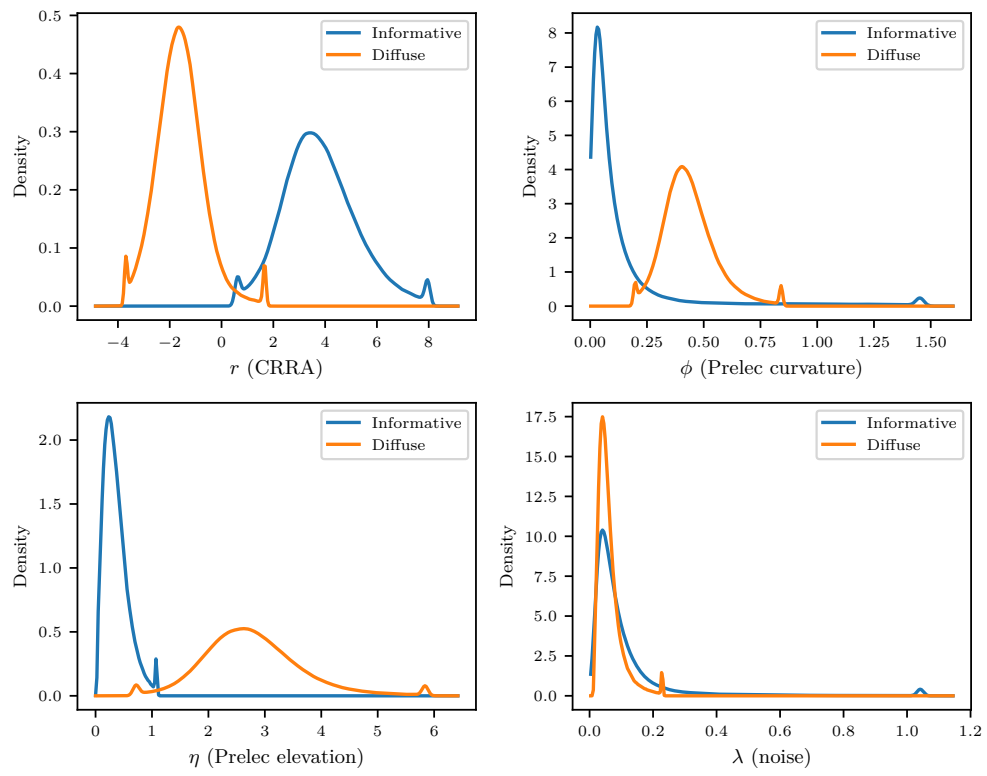
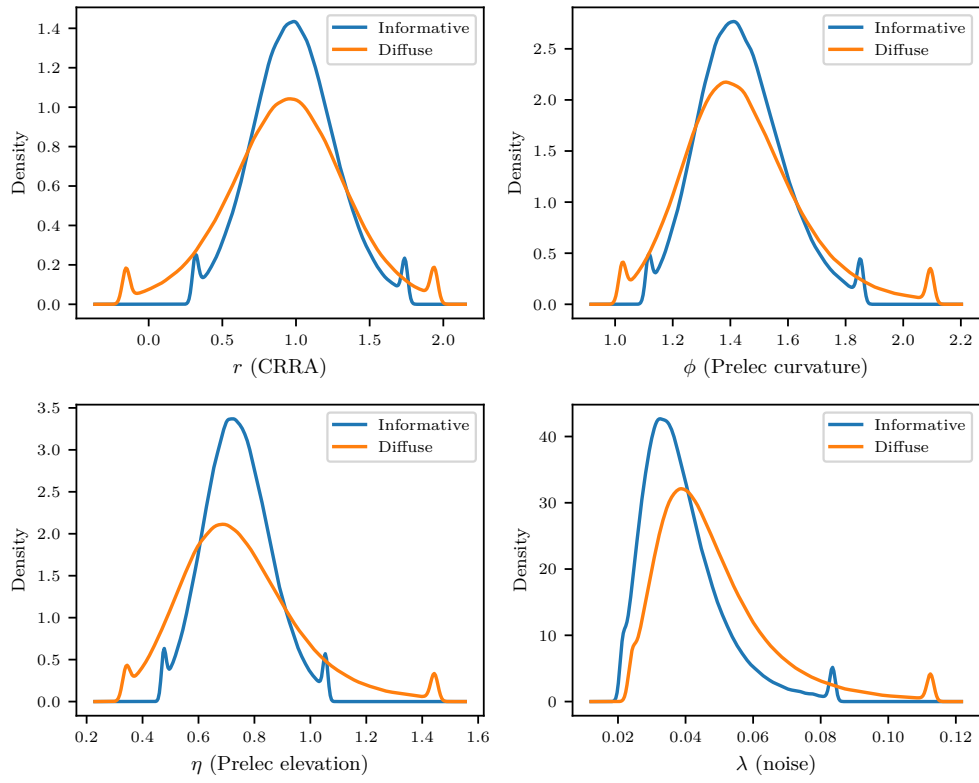


Figure D.9: Posterior parameter distributions: least affected subject



pattern is stable across the four configurations: the same subset of subjects remains prior-sensitive, the lottery-specific ordering of posterior predictive CEs is preserved, and the informative prior continues to act primarily as a regularizer where the data are least informative. In that sense, the substantive conclusions are not tied to a single baseline choice of (α, κ) , even though the magnitude of prior influence varies in the expected direction across configurations.

Appendix E: MCMC Estimation and Marginal Likelihood Details

All three models are estimated using a Metropolis-within-Gibbs sampler. The parameter vector is partitioned into blocks, with each subject’s structural parameters forming one block and each pair of hyperparameters (population mean and dispersion for each quantity) forming another. Each block is updated sequentially using a Metropolis-Hastings step with an adaptive multivariate normal proposal distribution. The proposal covariance is estimated from the empirical covariance of an A/B chain split: Group A chains use the running covariance of Group B to construct proposals, and vice versa. This coupled design improves mixing while maintaining detailed balance within each group.

For each model, 2000 chains are run in parallel on GPU (1000 per A/B group). Every 10th iteration is retained (thinning factor of 10), and the first half of the retained samples is discarded as burn-in. Table E.1 summarizes the sampling configuration and convergence diagnostics for each model.

All three models share the CRRA utility function, so their per-iteration cost is similar, governed by the analytic Jacobian decomposition for Models 1 and 2 and the Gram determinant $\sqrt{\det(J^\top J)}$ for Model 3. Model 3 is the fastest per iteration because it has the fewest structural parameters and uses the analytic CRRA inverse.

The split $\hat{R}$ statistic is computed for every subject-parameter combination

	Model 1 (CRRA + Prelec)	Model 2 (CRRA + GE)	Model 3 (CRRA + Power)
Structural parameters per subject	4	4	3
Predictive quantities	4	4	4
Hyperparameters	8	8	8
Total parameters	452	452	341
Chains (A + B)	2000	2000	2000
Collection iterations	6000	6000	10000
Thinning	10	10	10
Retained samples per chain	600	600	1000
Post burn-in samples per chain	300	300	500
Total posterior draws	600,000	600,000	1,000,000
Median $\hat{R}$	1.020	1.038	1.018
Max $\hat{R}$	6.76	6.51	8.94
Mean acceptance rate	50.0%	45.7%	49.5%
Time per iteration (s)	0.82	0.73	0.67

Table E.1: MCMC sampling configuration and convergence diagnostics.

and every hyperparameter, using the second half of the retained samples split again into two segments. The share of parameters with $\hat{R}$ above 1.1 is 14%, 23%, and 8% for Models 1, 2, and 3. Elevated values are concentrated in the noise parameter λ for subjects whose choice data are consistent with multiple modes in the posterior, and for such genuinely multimodal subject posteriors the split statistic overstates non-convergence, since chains that settle in different modes inflate the between-chain variance even at stationarity. These are the weakly identified subjects discussed in Section 5, whose choices are consistent with multiple risk preference profiles. The eight hyperparameters mix far better than the worst subject parameters, with maximum hyperparameter $\hat{R}$ of 1.15, 1.25, and 1.10 respectively.

E.1 Marginal Likelihood Computation

The marginal likelihoods reported in Section 5.4 exploit the conditional independence of subjects given the hyperparameters. The marginal likelihood factors as

$$p(y \mid M) = \int p(\boldsymbol{\gamma} \mid M) \prod_{i=1}^N p(y_i \mid \boldsymbol{\gamma}, M) d\boldsymbol{\gamma}, \quad (\text{E.1})$$

where each integrated likelihood $p(y_i \mid \boldsymbol{\gamma}, M)$ is an integral over the three or four structural parameters of subject i . Estimating the low-dimensional factors separately avoids the need for a proposal distribution that covers the full 452- or 341-dimensional parameter space, which is the binding practical constraint for direct bridge sampling in models of this size.

Each per-subject integral is estimated by importance sampling with $K = 16384$ fresh draws (32768 for Model 2) from a proposal tailored to that subject. The proposal is a mixture of Student- t components fitted to the subject’s marginal posterior draws, stratified along the first principal component so the mixture follows the curvature of the posterior, and refined once by importance resampling against the integrand evaluated at a spread of posterior hyperparameter draws. The refinement matters because the conditional posterior of a subject given $\boldsymbol{\gamma}$ is substantially narrower than the subject’s marginal posterior. The hyperparameter dependence of the Beta-Beta population density enters the importance weights only through the Beta coefficients, which multiply functions of the structural parameters that are cached once, so sweeping the per-subject estimates across thousands of hyperparameter draws costs two tensor contractions per draw.

The remaining 8-dimensional integral over $\boldsymbol{\gamma}$ in Equation (E.1) is computed

by two routes that share no integration logic. The first is importance sampling from a Student- t distribution fitted to the hyperparameter posterior. The second is the iterative bridge sampling estimator of Meng and Wong (1996) between the posterior draws and the same proposal. Because the per-subject estimates enter the integrand multiplicatively, a single set of importance draws yields a log marginal likelihood biased downward by approximately $\sum_i 1/(2 \text{ESS}_i)$, with standard errors blind to the shared draws. The procedure therefore runs 8 independent replicates with fresh draws, combines the replicate estimates on the natural scale, and takes standard errors from the replicate spread. The two routes agree to within 0.11 log units across the three models. The validity of the estimator does not rest on the mixing of the subject-level chains. Posterior draws of the structural parameters enter only as inputs to the proposal fit, where imperfect mixing costs efficiency rather than correctness, and the bridge route consumes hyperparameter draws, which are well mixed for all three models. The residual gap between the routes for Model 2 is consistent with its slightly elevated maximum hyperparameter $\hat{R}$.

The estimator was validated end to end on a synthetic dataset of 5 subjects making 20 choices each, generated from known parameters, where the marginal likelihood can be computed by brute force using draws taken exactly from the Beta-Beta hyperprior. Two brute-force runs with different inner proposals agree to within one hundredth of a log unit, and the estimator recovers their value, -72.38 , to within 0.03 log units on the importance sampling route and 0.03 on the bridge route.

E.2 Feasible-Mass Normalization

The values above integrate the prior as constructed, before the truncation normalization described in Section 5.4. The feasible mass $C_m(\boldsymbol{\gamma})$ is a single 3- or 4-dimensional integral per model and hyperparameter value, the same object as a per-subject integrated likelihood with the likelihood replaced by one, and is estimated by the same adaptive importance sampling machinery. Proposals are initialized from the model’s prior-only MCMC draws and refined against the integrand at a spread of hyperparameter values; effective sample sizes at the final stage are in the tens of thousands, and independent replicates agree to about one hundredth of a log unit at the posterior mean.

The normalizer $Z_m = \mathbb{E} [C_m(\boldsymbol{\gamma})^N \mid \boldsymbol{\xi}]$ is more delicate because the power $N = 111$ amplifies both the variation of C_m across $\boldsymbol{\gamma}$ and any estimation error, and because the tilted distribution $\pi(\boldsymbol{\gamma}) C_m(\boldsymbol{\gamma})^N$ concentrates along a curved ridge of hyperparameter values rather than at a point, which defeats mode-centered approximations. It is estimated by annealed adaptive importance sampling. The tilt exponent is raised from 0 to N in adaptive steps chosen to keep the incremental effective sample size above thirty percent. At each step fresh particles are drawn from a stratified Student- t mixture fitted to the previous step’s tilted cloud, the same ridge-following construction used for the per-subject proposals, and the feasible-mass estimator is rebuilt around the current cloud. The reported value is the direct importance sampling estimate at the full tilt from the final proposal, so intermediate steps guide the proposal without contributing telescoping error. Final effective sample sizes are around thirty percent of the particle budget, and independent replicates of the entire procedure agree to a small fraction of a

log unit. Two further checks corroborate the estimates. For Models 1 and 2 the hard bound $\log Z_m \leq N \max_{\gamma} \log C_m(\gamma)$ implied by $C_m \leq 1$ is respected, and a separate trust-region scheme whose hull truncation makes it a lower bound lands about two log units below the annealed value, consistent with the small mass it omits.

For Model 3 the annealing reliably reaches a tilt of roughly $t = 60$ and then encounters what appears to be a first-order transition: the tilted mass splits toward a small-heterogeneity corner of the hyperparameter space where the surface integral C_3 grows, and independent runs stall at the same tilt rather than crossing. Instead of an exact Z_3 , the paper therefore reports a rigorous lower bound. The function $t \mapsto \log Z(t) = \log \mathbb{E}[C(\gamma)^t \mid \xi]$ is convex, so for any $t^* < N$,

$$\log Z(N) \geq \log Z(t^*) + (N - t^*) \mathbb{E}_{t^*}[\log C], \quad (\text{E.2})$$

where $\mathbb{E}_{t^*}$ is taken under the tilted distribution at t^* . Both quantities on the right are computed at a healthy annealing stage well before the stall ($t^* = 40$, effective sample sizes around 20,000, two replicates agreeing to two tenths of a log unit), giving $\log Z_3 \geq -100.8$. The resulting normalized log marginal likelihood for Model 3 is at most -5325.6, so the log Bayes factor is at least 107.4 for Model 1 over Model 3 and at least 91.3 for Model 2 over Model 3, the lower bounds quoted in Section 5.4. The numerical difficulty itself is informative: for $k > p$ the unnormalized induced construction tilts the effective hyperprior toward extreme heterogeneity values, a caveat for over-determined quantity-space priors that Section 6 takes up.